

Long Rates, Life Insurers, and Credit Spreads*

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Abstract

This paper proposes a new channel through which long-term interest rates transmit to credit spreads. When life insurers carry negative duration gaps, higher rates reduce their liabilities more than assets. Rate increases therefore boost equity and risk-bearing capacity, lowering equilibrium credit spreads. Empirically, I test this channel with bond-level yields and a maturity-based discontinuity in bond ownership. Insurers' trades confirm the mechanism: after rates rise, insurers shift portfolios toward riskier, high-yield bonds. As rates increase, bonds more heavily held by life insurers experience greater spread reductions. The results show that institutional duration mismatch shapes credit spreads and corporate financing conditions.

JEL Codes: G12, G22, G23, E50

Keywords: Credit Spreads, Long-term Interest Rates, Life Insurers, Interest Rate Risk, Monetary Policy

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The U.S. corporate bond market has expanded dramatically over the last four decades, with total outstanding reaching 56% of GDP in 2023, up from 17% of GDP in 1983, and surpassing the bank loan market as the primary funding source for U.S. firms. Over the past 40 years, the U.S. corporate bond market has been dominated by large institutional investors. Among them, life insurance companies have consistently been the largest investor group, holding over 20% of outstanding corporate bonds. Given their dominant position in the bond market, life insurers' demand can have a considerable impact on bond prices and credit spreads.

How do life insurers affect bond pricing? The existing literature has identified two potential channels. First, insurers face risk-based capital regulations, so negative shocks to their capital or the capital charges of bonds restrict their bond demand, resulting in fire sales and fluctuations in credit spreads (Ellul, Jotikasthira, and Lundblad 2011; Murray and Nikolova 2022). Second, the liabilities of life insurers mostly consist of life insurance and annuity products, which are inherently long-term and illiquid. Hence, insurers have more stable funding than most other institutional bond investors, and they act as a stabilizing force in times of market distress (Chodorow-Reich, Ghent, and Haddad 2021; Coppola 2025). In this paper, I provide evidence in support of a new duration mismatch channel where interest rates influence insurers' bond demand and equilibrium credit spreads through life insurers' interest rate risk exposure.

Modern life insurers issue long-duration products and invest the proceeds in fixed income markets. The long-lived liabilities naturally expose life insurers to interest rate risk, as higher interest rates, especially higher long-term interest rates, compress insurers' liabilities more than their assets and generate capital gains. As shown in the earlier literature (e.g., Koijen and Yogo 2022; Sen 2023), U.S. life insurers have sustained large negative duration gaps since the 2007–2008 Global Financial Crisis (GFC). Post-GFC, increases in long rates strengthen life insurers' capital and expand their risk-taking capacity. Consequently, insurers might increase their demand for riskier bonds and rebalance their portfolios towards them. Due to the large share of life insurers in the bond market, such demand shifts can significantly reduce credit spreads.

I first develop a tractable model of the corporate bond market featuring duration-mismatched life insurers as key investors to illustrate the mechanism. In the model, life insurers hold corporate bonds and Treasuries as assets and issue long-term annuities as liabilities. Life insurers face a duration mismatch because their liabilities have a much longer duration than their assets. Increases in long-term Treasury yields lead to larger declines in liabilities' value than in assets' value, boosting insurers' net worth and

risk-bearing capacity. Thus, when the long rate rises, life insurers become more willing to hold bonds with lower credit ratings, which carry greater credit risk and regulatory costs. The increase in insurers' bond demand then leads to lower equilibrium credit spreads. Therefore, the model implies that negative duration gaps of life insurers cause a negative aggregate comovement between long rates and credit spreads.

Following the model predictions, I document novel facts about the comovements between long-term interest rates and corporate bond credit spreads over time. Leveraging detailed microdata on corporate bond prices, I estimate the pass-through of the 10-year Treasury yield to corporate bonds with different credit ratings. Before the Financial Crisis, when life insurers had no significant interest rate risk exposure, long rates and credit spreads exhibited no significant associations. Changes in the long-term interest rate affected all bonds in a similar fashion. However, after the Financial Crisis, as life insurers faced severe negative duration gaps, an economically and statistically significant negative relationship between long rates and credit spreads emerged. In particular, yields on low-credit-rating bonds decline relative to those on high-credit-rating bonds when long-term interest rates rise. In fact, yields on high-yield bonds exhibit statistically significant declines in absolute terms when the long-term interest rate increases. In addition, I find that the negative comovements become more pronounced when life insurers' duration gap is more negative.

Contrary to the conventional wisdom where high interest rates raise risk premia by increasing default risk (as in standard corporate finance models, e.g., [Hennessy and Whited 2007](#); [Gomes and Schmid 2010](#)) and discouraging risk-taking (e.g., [Jiménez et al. 2014](#); [Bauer, Bernanke, and Milstein 2023](#)), increases in the long-term interest rate are followed by declines in corporate bond credit spreads, even after controlling for changes in credit risk and detailed bond characteristics. Moreover, the impact is more significant for bonds with lower credit ratings: when the long-term interest rate declines, the credit spreads of high-yield bonds increase by significantly more than those of investment-grade bonds. Overall, the evidence suggests that an alternative mechanism, such as the duration mismatch of life insurers, might be at play.

High-frequency identifications around monetary policy announcements further demonstrate that increases in long rates reduce corporate bond credit spreads. I construct high-frequency shocks to the 10-year Treasury yield using yield movements around Federal Open Market Committee (FOMC) meetings. Using the local projection method, I find that 10-year Treasury yield shocks did not cause significant changes in bond credit spreads before the Financial Crisis. After the Crisis, positive shocks to long-term

Treasury yields induce large negative responses in credit spreads. For example, a 1% increase in the 10-year Treasury yield around FOMC meetings leads to an almost 1% reduction in the spread between single B corporate bonds and AAA corporate bonds.

Next, I trace this negative comovement to life insurers, the largest institutional investor group in the U.S. bond market, which own over 20% of all U.S. corporate bonds. The business model of modern life insurers involves investing in fixed-income securities and issuing long-term life insurance and annuity products. Before the Financial Crisis, life insurers were hedged against interest rate risk. Their equity value was largely shielded from fluctuations in the long-term interest rate, suggesting that their assets and liabilities had matching durations. However, as the economy entered a low-interest-rate environment after 2008, life insurers began to face a severe duration mismatch, and their market equity values became highly sensitive to long-term interest rates.¹ In this time period, the market equity of the life insurance sector increased by more than 6% when the 10-year Treasury yield rose by 1%.

I provide causal evidence that greater life insurance ownership generates stronger comovement between long-term interest rates and credit spreads. To sharpen the identification, I exploit a discontinuity in bond ownership structure stemming from mutual funds' investment mandates (e.g., [Li and Yu 2026](#); [Bretscher, Schmid, and Ye 2023](#)). [Figure 1](#) visualizes the composition of the U.S. bond market. The bond market is populated by large institutional investors, with life insurers as the largest investor type, followed by mutual funds. Among bond mutual funds, many are “intermediate-term” with fund charters and mandates to invest in bonds whose maturities are less than 10 years, resulting in a discontinuity in investor composition around the 10-year maturity threshold. Bonds with maturities slightly below 10 years are significantly less likely to be held by life insurers than bonds with maturities slightly above 10 years, as the former face higher demand from mutual funds. I then find that bonds with time to maturity slightly below 10 years are much less responsive to the long rate.

Again, the results are consistent with a new channel through which the long-term interest rate and monetary policy affect the corporate bond market—the duration mismatch channel. Life insurers' net worth rises following increases in the long-term interest rate. Consequently, their risk-bearing capacity is higher, and they expand their holdings

¹Lower interest rates increase the insurers' liability duration more than their asset duration. Together with other institutional frictions (e.g., regulations, transaction costs, and market incompleteness), it can leave insurers in a persistent duration mismatch. See [Section 4.1](#) for detailed discussions on the change in life insurers' hedging behavior.

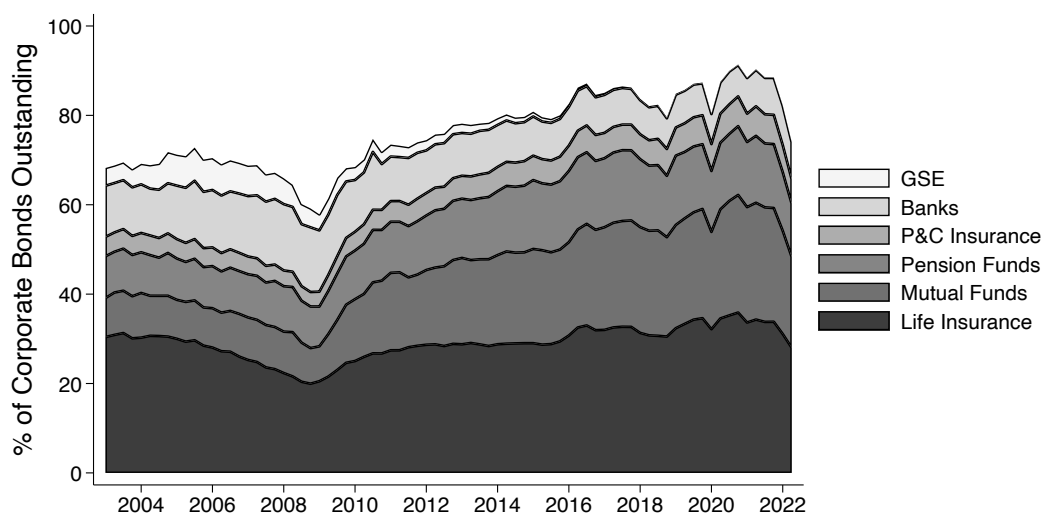


Figure 1. Institutional investors in the bond market

This figure plots the ownership structure of the U.S. corporate and foreign bond market between 2004 and 2022 (excluding holdings by foreign investors). From top to bottom, the figure shows the fraction of bonds owned by government-sponsored enterprises, commercial banks, property-casualty insurance companies, pension funds, mutual funds, and life insurance companies. The data are obtained from the Financial Accounts of the United States.

of risky bonds. Furthermore, I show that life insurers' bond trading behavior supports the duration mismatch channel. Following increases in the 10-year interest rate, life insurers respond by increasing their demand for risky bonds, as evidenced by more future purchases of risky corporate bonds.

Related Literature My results contribute to the extensive literature on corporate bond credit spreads.² Following [Fama and French \(1993\)](#), many have modeled corporate bond yields and returns using factor models (e.g., [Gebhardt, Hvidkjaer, and Swaminathan 2005](#); [Lin, Wang, and Wu 2011](#); [Acharya, Amihud, and Bharath 2013](#); [Jostova et al. 2013](#); [Kelly, Palhares, and Pruitt 2023](#)). The factor approach typically views credit risk and the term structure as unconnected orthogonal factors in determining bond yields. In this paper, I show that the pricing of the two factors is interconnected, as the level of long-term interest rates can affect credit risk pricing through the risk-bearing capacity of life insurers. Another strand of literature aims to explain the levels and fluctuations of corporate bond credit spreads (see, e.g., recent works by [Eom, Helwege, and Huang 2004](#); [Longstaff, Mithal, and Neis 2005](#); [Schaefer and Strebulaev 2008](#); [Chen, Collin-](#)

²See [Huang and Shi \(2021\)](#) for an overview of the literature on corporate bond returns.

Dufresne, and Goldstein 2009; Huang and Huang 2012; Kuehn and Schmid 2014; Culp, Nozawa, and Veronesi 2018; Feldhütter and Schaefer 2018; van Binsbergen, Nozawa, and Schwert 2025). In particular, Longstaff and Schwartz (1995) also finds a negative relationship between Treasury yields and the bond-Treasury spreads and emphasizes the role of changes in the risk-neutral default probability. Similarly, Duffee (1998) finds a negative comovement between term spreads and spreads between corporate bonds and Treasuries, highlighting the role of corporate bond callability. In this paper, I show that the negative relationship between credit spreads and long rates post-2008 is not explained by the market price of default risk, measured by Merton's expected default frequency (EDF), or the moneyness of call options. I contribute to this literature by proposing a new mechanism that drives the negative relationship between credit spreads and long-term interest rates, centered on the interest rate risk exposure of life insurance companies.

This paper also belongs to the burgeoning literature focusing on the role of institutional investors in the corporate bond market. My work is most closely connected to Coppola (2025) and Li and Yu (2022, 2026), who, in different contexts, also show that investor composition matters for bond price dynamics. Using similar empirical frameworks, I show that life insurance ownership induces a negative comovement between credit spreads and the long-term interest rate after the Financial Crisis. A recent strand of the literature studies the role of mutual funds in liquidity disruptions in crises (e.g., Haddad, Moreira, and Muir 2021; Falato, Goldstein, and Hortaçsu 2021; Jiang et al. 2022; Ma, Xiao, and Zeng 2022). This paper, instead, focuses on life insurers, the largest investor group in the U.S. corporate bond market, which currently holds more than 20% of the U.S. corporate market capitalization, to show that their balance sheets are important for bond price dynamics. Recent works such as Bretscher et al. (2026) and Darmouni, Siani, and Xiao (Forthcoming) use demand system approaches to analyze the equilibrium effects of institutional demand. In this paper, I depart from the typical demand functions used in the demand system literature to analyze the role of life insurers' duration mismatch.

The findings of my paper echo recent research on the investing behavior of life insurance companies (e.g., Koijen and Yogo 2022, 2023; Chodorow-Reich, Ghent, and Haddad 2021; Ellul et al. 2022). Several studies, including Berends et al. (2013), Hartley, Paulson, and Rosen (2016), Domanski, Shin, and Sushko (2017), Ozdagli and Wang (2019), Koijen and Yogo (2022), Huber (2022), Sen (2023), Kirti and Singh (2025), and Li and Wenning (2025) have also shown that life insurers' interest rate risk exposure

changed after the Financial Crisis.³ I build on this finding and argue that the duration mismatch can potentially have a large influence on bond prices since life insurers, on average, hold more than 20% of U.S. corporate bonds. Prior works by [Ellul, Jotikasthira, and Lundblad \(2011\)](#), [Nanda, Wu, and Zhou \(2019\)](#), [Girardi et al. \(2021\)](#), [Becker, Opp, and Saidi \(2022\)](#), and [Murray and Nikolova \(2022\)](#) have established that the trades of life insurers have potentially large price impacts on corporate bond prices. I confirm that life insurers can considerably influence bond prices and propose a new mechanism stemming from their interest rate risk. Consistent with [Ge and Weisbach \(2021\)](#) and [Bhardwaj, Ge, and Mukherjee \(2022\)](#), I find that life insurers tilt their portfolios towards safer investments when they become financially constrained.

The approach of this paper connects to the literature on intermediary asset pricing, which emphasizes the role of the financial health of intermediaries on asset prices (e.g., [Brunnermeier and Pedersen 2009](#); [He and Krishnamurthy 2013, 2018](#); [Brunnermeier and Sannikov 2014](#); [Adrian, Etula, and Muir 2014](#); [He, Kelly, and Manela 2017](#); [Haddad and Muir 2021](#); [Baron and Muir 2022](#)). My paper finds that the duration mismatch of life insurers, the dominant intermediaries in the U.S. corporate bond market, significantly affects bond credit spreads. Unlike previous works, which find that higher interest rates depress intermediaries' net worth (e.g., [Gomez et al. 2021](#); [Kekre, Lenel, and Mainardi 2023](#)), I find that life insurers face the opposite duration mismatch and experience equity gains when long rates increase.

Additionally, this paper contributes to the literature on the effect of monetary policy and interest rates on the bond market. Departing from existing works that focus on short-term monetary policy and mutual funds (e.g., [Guo, Kontonikas, and Maio 2020](#); [Daniel, Garlappi, and Xiao 2021](#); [Chen and Choi 2024](#); [Fang 2023](#)), I instead focus on the long-term interest rate and life insurers. I document evidence that positive shocks to long-term interest rates significantly depress corporate bond credit spreads and encourage bond issuance by risky firms.

Another strand of the literature on the comovement between Treasury and stock returns focuses more on inflation and output dynamics. For example, [Campbell, Pflueger, and Viceira \(2020\)](#) find that the Treasury-stock comovement turned negative around 2001 and that periods of high Treasury returns tend to coincide with periods of low output that hurt stock returns.⁴ In this paper, I find that the negative comovement between long rates and credit spreads emerged much later than the negative Treasury-stock

³[Li and Wenning \(2025\)](#) study the product market implications of life insurers' duration mismatch.

⁴See also [Baele, Bekaert, and Inghelbrecht \(2010\)](#), [David and Veronesi \(2013\)](#), and [Song \(2017\)](#).

Assets	Liabilities
Corporate Bonds	Long-Duration Annuities
Treasuries	Net Worth

comovement and is driven mostly by the risk exposure of life insurers instead of default risk.

1 An Intermediary Asset Pricing Model

In this section, I build an intermediary asset pricing model centered around life insurers to illustrate how long rates affect insurers' bond demand and credit spreads. As in [He and Krishnamurthy \(2013\)](#), the risk-bearing capacity of intermediaries (life insurers) is key to equilibrium credit spreads. As in [Koijen and Yogo \(2023\)](#), life insurers take on leverage by issuing annuity liabilities and investing in corporate bonds. Importantly, life insurers' portfolios expose them to duration mismatch (see [Section 4.1](#) for empirical evidence), and their net worth increases with the long-term interest rate. The equity gains reduce life insurers' effective risk aversion and increase their risk-bearing capacity, which leads to lower credit spreads in equilibrium.

1.1 Model setup

Investors There are two types of corporate bond investors in the model—life insurers and preferred-habitat investors.

The life insurer invests its portfolio in corporate bonds and Treasuries while issuing long-term annuities to households. The duration mismatch between the insurer's assets (corporate bonds and Treasuries) and liabilities (annuities) exposes it to interest rate risk. When the interest rate on long-term Treasuries rises, the insurer's liabilities decline more than its assets, thereby increasing its net worth. The insurer's balance sheets are as follows.

Preferred-habitat investors include all other investors of corporate bonds (e.g., mutual funds). For simplicity, their demand for corporate bonds is assumed to be reduced-form functions of bond prices, in the spirit of [Vayanos and Vila \(2021\)](#).

Treasuries and Annuities Time is continuous. Treasuries are long-term consol bonds with a geometric maturity structure. The Treasuries have a price P_t^T , coupon rate ϕ^T , and a geometric decaying rate λ^T . Specifically, each bond pays a stream of coupon payments $\phi^T dt$ and has a face value that decays over time at a constant rate of λ^T . The return rate on Treasuries is

$$dr_t^T = \frac{(\phi^T - \lambda^T P_t^T)dt}{P_t^T} + \frac{dP_t^T}{P_t^T}, \quad (1)$$

where $\phi^T dt$ captures the coupon payments, $\lambda^T P_t^T dt$ captures the depreciated face value, and dP_t^T is the capital gain from price fluctuations. Based on the geometric decaying maturity structure, the average maturity of this bond is $\tau^T = 1/\lambda^T$. Each unit of the face value is expected to last a period of length τ^T . I define the Treasury yield as⁵

$$y_t^T := \frac{\phi^T}{P_t^T} - \lambda^T.$$

The Treasury yield has an exogenous law of motion and always reverts to the “natural” level $\bar{y}^T$ at a speed of α_y . I assume that the dynamics of the long-term Treasury yield are driven by (unspecified) monetary policy and are exogenous to the model.

Annuities are similar consol bonds with a price P_t^L , coupon rate ϕ^L , and a geometric decaying rate λ^L . Annuities have a maturity of $\tau^L = 1/\lambda^L$ and a return rate of

$$dr_t^L = \frac{(\phi^L - \lambda^L P_t^L)dt}{P_t^L} + \frac{dP_t^L}{P_t^L}.$$

Similar to Treasuries, I define the annuity yield as $y_t^L := \phi^L / P_t^L - \lambda^L$.

No-Arbitrage Pricing of Annuities In this model, both Treasuries and annuities are risk-free assets with deterministic returns, and the life insurer holds both in equilibrium. Therefore, the returns of Treasuries and annuities must be equalized,

$$\mu_t^{r,T} = \mu_t^{r,L}, \quad (2)$$

⁵The yield to maturity is defined as the rate at which future payoffs are discounted and aggregated into the current bond price:

$$P_t^T = \int_t^\infty e^{-(y_t^T + \lambda^T)s} \phi^T ds.$$

where $\mu_t^{r,T} = dr_t^T/dt$ and $\mu_t^{r,L} = dr_t^L/dt$ are the return rates of Treasuries and annuities, respectively. This no-arbitrage condition allows us to solve for the annuity price P_t^L and yield y_t^L as functions of the Treasury yield y_t^T .

Corporate Bonds Corporate bonds are long-term bonds with credit risk. There are N types of corporate bonds, each corresponding to a credit rating. Exposure to credit risk varies across ratings. Investors hold diversified portfolios within each rating. The return rate on rating $n \in \{1, \dots, N\}$ bonds is

$$dr_t^n = \frac{\phi^n - \lambda^n P_t^n}{P_t^n} dt + \frac{dP_t^n - \nu^n P_t^n dJ_t}{P_t^n}. \quad (3)$$

The coupon rate is denoted by ϕ^n , while the rate of decay is represented by λ^n . The process dJ_t is a Poisson jump process with intensity δ that captures the bonds' credit risk.⁶ When a jump is realized, the bond market enters a disrupted period, and a fraction ν^n of rating n bonds default and lose their value. As before, the corporate bond yields are defined as $y^n := \phi^n / P_t^n - \lambda^n$. Further, I define

$$dr_t^n = \mu_t^{r,n} dt + \sigma_t^{r,n} (dJ_t - \delta dt),$$

so that $\mu_t^{r,n}$ is the expected return on bonds n and $\sigma_t^{r,n}$ is the overall risk exposure of bonds n .⁷

Life Insurer's Portfolio Problem There is one representative life insurer that holds corporate bonds and Treasuries while issuing annuities to households. The quantity of annuities is assumed to be exogenous at L ,⁸ and the insurer chooses its holdings of corporate bonds and Treasuries. Its portfolio weights in Treasuries, annuities, and corporate bonds are denoted by $w_t^{I,T}$, $w_t^{I,L}$, and $\left(w_t^{I,n}\right)_{n=1}^N$, respectively. The insurer's

⁶The credit risk is correlated across different bonds. This is consistent with the findings of [Das et al. \(2007\)](#) and [Duffie et al. \(2009\)](#), who show that corporate bond defaults in the United States are highly correlated.

⁷The process $(dJ_t - \delta dt)$ is a martingale since the process dJ_t drifts upwards in expectation ($\mathbb{E}_t[dJ_t] = \delta dt$).

⁸In practice, the quantity of life insurers' liabilities is stable as they are typically long-term products with few early withdrawals (e.g., [Chodorow-Reich, Ghent, and Haddad 2021](#)).

net worth A_t^I evolves according to

$$\frac{dA_t^I}{A_t^I} = \left[w_t^{I,T} \mu_t^{r,T} + \sum_{n=1}^N w_t^{I,n} \mu_t^{r,n} - w_t^{I,L} \mu_t^{r,L} - \sum_{n=1}^N \frac{1}{2} \left(\zeta^n w_t^{I,n} \right)^2 + \xi_t \right] dt + \sum_{n=1}^N w_t^{I,n} \sigma_t^{r,n} dJ_t.$$

$w_t^{I,L} \mu_t^{r,L}$ enters the net worth evolution negatively because annuities are liabilities. The insurer incurs regulatory costs $\frac{1}{2} \sum_{n=1}^N \left(\zeta^n w_t^{I,n} \right)^2 A_t^I$ for holding corporate bonds, which potentially include the costs of complying with risk-weighted leverage constraints and passing stress tests.⁹ Following [Kojen and Yogo \(2023\)](#), the regulatory cost is assumed to be quadratic in the bond holdings, while the parameter ζ^n governs the marginal regulatory cost of holding bonds in rating n . The process $\xi_t dt$ captures an exogenous process of equity injections and dividend payouts, ensuring that net worth A_t^I is stationary. The insurer has a mean-variance preference over its growth rate:

$$\mathbb{E}_t \left[\frac{dA_t^I}{A_t^I} \right] - \frac{a}{2} \text{Var}_t \left[\frac{dA_t^I}{A_t^I} \right].$$

The portfolio problem can then be written as¹⁰

$$\max_{\{w_t^{I,n}\}_{n=1}^N, w_t^{I,T}, w_t^{I,L}} \underbrace{w_t^{I,T} \mu_t^{r,T} + \sum_{n=1}^N w_t^{I,n} \mu_t^{r,n} - w_t^{I,L} \mu_t^{r,L}}_{\text{expected return}} - \underbrace{\frac{a}{2} \delta \left(\sum_{n=1}^N w_t^{I,n} \sigma_t^{r,n} \right)^2}_{\text{risk aversion}} - \underbrace{\frac{1}{2} \sum_{n=1}^N \left(\zeta^n w_t^{I,n} \right)^2}_{\text{regulatory cost}}, \quad (4)$$

$$\text{s.t.} \quad w_t^{I,T} + \sum_{n=1}^N w_t^{I,n} - w_t^{I,L} = 1. \quad (5)$$

The objective function (4) can be decomposed into three components: expected portfolio return, exposure to credit risk, and additional regulatory costs. The parameter a symbolizes the degree of risk aversion. Equation (5) is the insurer's balance sheet constraint, which simply states that the portfolio weight in assets (Treasures and corporate bonds) minus the portfolio weight in liabilities (annuities) equals one, the portfolio weight in net worth.

⁹Life insurers also face other potential costs, such as transaction and informational costs, which are conceptually similar to the regulatory cost from the modeling perspective.

¹⁰It follows from the fact that the $\text{Var}_t(dJ_t) = \delta dt$.

Preferred-Habitat Investors and Bond Supply The preferred-habitat investors also participate in the corporate bond market. I denote the shares of rating n bonds held by the preferred-habitat investors as $D_t^{P,n}$. I assume that the habitat demand is a reduced-form function of bond prices $D_t^{P,n} = D^{P,n}(P_t^n)$. The preferred-habitat demand is downward-sloping, that is, $(D^{P,n})'(\cdot) \leq 0$.¹¹

There exists a firm sector that supplies corporate bonds. For simplicity, I model bond supply as a function of bond prices, $B_t^n = B^n(P_t^n)$. The supply function is naturally upward-sloping (i.e., $(B^n)'(\cdot) \geq 0$) because firms face lower borrowing costs when their bond prices are higher, and they exploit the lower borrowing costs by issuing additional bonds.¹²

Market Clearing There are three asset markets in this model: Treasury, corporate bond, and annuity. Recall that the Treasury yield is exogenously given. Treasuries are supplied perfectly elastically to clear the market, given the exogenous yield. The market for rating n corporate bonds clears when

$$w_t^{I,n} A_t^I + D_t^{P,n} = B_t^n. \quad (6)$$

The market-clearing condition requires that the sum of the life insurer's demand $w_t^{I,n} A_t^I$ and preferred-habitat investors' demand $D_t^{P,n}$ is equal to the total market cap B_t^n . Finally, the annuity market clears when the insurer's portfolio weight in annuities $w_t^{I,L}$ matches the exogenously fixed annuity demand,

$$w_t^{I,L} A_t^I = P_t^L L. \quad (7)$$

1.2 Analytical insights

In this section, I analyze the effects of an unexpected shock to the long-term Treasury yield on credit spreads. I focus on a simplified version of the model that permits analytical solutions and illustrates the mechanism. To this end, I make two simplifying assumptions.

Assumption 1. *Corporate bonds are short-term bonds with independent default risks.*

¹¹The assumption is standard in the literature of institutional asset demand and supported by various empirical estimates (e.g., [Bretscher et al. 2026](#); [Darmouni, Siani, and Xiao Forthcoming](#)).

¹²I derive a micro-founded form of the supply function in Internet Appendix D.

Since the corporate bonds are short-term with independent credit risks, their return rates (3) simply become

$$dr_t^n = \frac{\phi^n}{P_t^n} dt - v^n dJ_t^n,$$

where $\{dJ_t^n\}$ are independent from each other. To understand how the insurer's net worth affects bond credit spreads, it is useful to characterize the solution to the portfolio problem (4)-(5). In this simplified model, the first-order condition for rating n bonds is given by

$$s_t^n := \underbrace{\mu_t^{r,n} - \mu_t^T}_{\text{credit spread}} = \underbrace{a\delta w_t^{I,n}(v^n)^2}_{\text{credit risk}} + \underbrace{(\zeta^n)^2 w_t^{I,n}}_{\text{regulatory premium}} = \gamma^n w_t^{I,n}, \quad (8)$$

where $\mu_t^{r,n} := \mathbb{E}_t[dr_t^n]/dt = \phi^n/P_t^n - v^n\delta$ is the expected return on rating n bonds and the constant $\gamma^n := a\delta(v^n)^2 + (\zeta^n)^2 > 0$ measures the insurer's total cost of holding rating n bonds. We can then write Equation (8) as

$$w_t^{I,n} = \frac{s_t^n}{\gamma^n}, \quad (9)$$

which is the standard portfolio choice condition under mean-variance preferences; $w_t^{I,n}$, the insurer's portfolio weight in rating n bonds, equals the credit spread s_t^n divided by the cost parameter γ^n .¹³

Equation (9) allows us to express $A_t^I w_t^{I,n}$, the (dollar) amount of corporate rating n bonds demanded by the life insurer, as

$$A_t^I w_t^{I,n} = \frac{s_t^n}{\gamma^n / A_t^I}.$$

The demand is given by the credit spread divided by the insurer's effective risk aversion, which I define as γ^n / A_t^I , the total holding cost γ^n relative to the net worth A_t^I . Under duration mismatch, an increase in Treasury yields boosts the insurer's net worth, thereby lowering its effective risk aversion. As a result, the insurer increases its demand for risky bonds, putting downward pressure on equilibrium credit spreads.

Assumption 2. *The habitat demand and bond supply functions take the forms of $D_t^{P,n} =$*

¹³In the absence of the regulatory cost ζ^n , the parameter $\gamma^n = a\delta v^n$ equals the risk aversion coefficient a times the default risk δv^n , which is exactly the solution of a standard mean-variance portfolio problem.

$\alpha^P + \beta^P s_t^n, B_t^n = \alpha^B - \beta^B s_t^n$. Hence, the net supply of rating- n bonds is a linear function of its credit spread, $B_t^n - D_t^{P,n} = \alpha - \beta s_t^n$, where $\alpha = \alpha^B - \alpha^P > 0, \beta = \beta^B + \beta^P > 0$.

This assumption imposes a tractable form for bond supply and habitat demand functions. When the price of a bond falls, it becomes cheaper relative to Treasuries and its credit spread widens. As we have discussed earlier, habitat investors increase their holdings when a bond becomes cheaper because their demand is downward-sloping. At the same time, firms reduce issuance as lower bond prices increase their borrowing costs. Therefore, higher credit spreads are accompanied by greater habitat demand and less bond supply, which leaves the insurer with a smaller net supply. For tractability, I assume that the net supply function is homogeneous across credit ratings. In Internet Appendix D, I relax this assumption and calibrate the habitat demand and bond supply to the empirical moments.

Proposition 1. *Suppose that the life insurer's net worth is sufficiently large, and it has a negative duration gap (i.e., $\partial A_t^I / \partial y_t^T > 0$). The following predictions hold when Treasury yields increase.*

1. *The credit spread s_t^n declines for all n .*
2. *The credit spread declines more for lower-quality bonds,*

$$\left| \frac{\partial s_t^n}{\partial y_t^T} \right| > \left| \frac{\partial s_t^m}{\partial y_t^T} \right| \quad \text{if } \gamma_n > \gamma_m.$$

3. *The supply of bonds shifts towards lower-quality bonds*

$$\frac{\partial (B_t^n - B_t^m)}{\partial y_t^T} > 0 \quad \text{if } \gamma_n > \gamma_m.$$

4. *The insurer rebalances towards lower-quality bonds,*

$$\frac{\partial (A_t^I w_t^{I,n} - A_t^I w_t^{I,m})}{\partial y_t^T} > 0 \quad \text{if } \gamma_n > \gamma_m.$$

5. *The magnitudes of predictions 1, 3, and 4 increase in the severity of the duration mismatch.*

The proof is relegated to [Appendix C](#). Proposition 1 summarizes the main analytical results from this model. When the yield on long-term Treasuries rises, the insurer's net

worth rises as it faces a duration mismatch, which reduces its effective risk aversion. The equilibrium credit spreads decline, with the effect stronger for lower-quality bonds. At the same time, the insurer rebalances towards lower-quality bonds. Moreover, the effect of higher interest rates is greater when the insurer's duration mismatch is more severe (i.e., when its net worth is more sensitive to Treasury yields).

1.3 Model extensions

Illustration of Quantitative Relevance While the model presented so far is primarily analytical, I quantify the model in Internet Appendix D and use a simple calibration to illustrate the quantitative importance of the channel. In the extension, I assume explicit functional forms for the bond supply function and the habitat demand. I then match key model parameters, including the life insurer's preferences, the bond market share of life insurers and habitat investors, and the habitat investors' demand elasticity, to existing estimates and empirical moments. The results show that the duration mismatch channel can potentially lead to sizable fluctuations in credit spreads, bond supply, and bond market composition following changes in the long-term interest rate.

Duration Management Motives The existing literature has also pointed to a duration management channel, where changes in interest rates affect insurers' demand for duration (e.g., [Ozdagli and Wang 2019](#); [Yu 2020](#)). The duration management channel suggests that insurers tilt their bond portfolios towards longer-duration bonds when low interest rates worsen their duration gaps.

In Internet Appendix E, I present an extension of the model that includes the duration management motives of life insurers to examine whether and how the duration mismatch channel proposed in this paper interacts with the duration management channel. In the extended model, life insurers first choose between bonds with different durations ("long-term vs. short-term") and then choose between bonds with different credit ratings ("risky vs. safe") within each duration.

Analytically, I show that the duration management channel governs the first portfolio choice problem, while the duration mismatch channel governs the second portfolio choice problem. Indeed, the life insurer rebalances its portfolio towards longer-duration bonds following interest rate decreases. However, as shown by [Domanski, Shin, and Sushko \(2017\)](#) and [Ozdagli and Wang \(2019\)](#), the reaching-for-duration behavior cannot entirely close the duration gap. Hence, the insurer still faces a negative duration gap in

equilibrium, and declines in interest rates continue to erode its net worth. Accordingly, in the second portfolio choice problem, the insurer further rebalances its portfolio towards safer bonds, within each duration.

In summary, through this model extension, I show that the effects of the duration management channel and the duration mismatch channel are separate: the former determines the insurer’s asset allocations across durations, while the latter determines the credit spread within each duration. If we focus on bonds of the same duration, the duration mismatch channel predicts that higher interest rates result in lower credit spreads for any given duration.

2 Data

In this paper, I combine data from multiple sources to assess the comovement between the long-term interest rate and corporate bond credit spreads, the significance of life insurers, and their real effects.

The Long-Term Interest Rate I use the Treasury yield curve at a daily frequency constructed by [Liu and Wu \(2021\)](#). In particular, I use the yield on 10-year U.S. Treasury notes as the proxy for the long-term interest rate.

Corporate Bonds I combine monthly data from the Mergent Fixed Income Securities Database (Mergent FISD), the Trade Reporting and Compliance Engine (TRACE), and the WRDS Bond Returns for corporate bond prices, yields, outstanding amounts, issuance, and other characteristics. The data set provides comprehensive coverage of publicly traded U.S. corporate bonds from 2002 to 2022.¹⁴ From Mergent FISD, I obtain information on bond issuers, maturities, durations, credit ratings, outstanding amounts, issuance dates, coupons, transaction volumes, and default histories at a monthly frequency. The WRDS Bond Returns data set also provides end-of-month transaction prices and yields extracted from TRACE. For any given month, I focus on bonds with at least one observed transaction price.

Mergent FISD reports three “raw” credit ratings from Standard and Poor’s (S&P), Moody’s Analytics, and the Financial Industry Regulatory Authority. For my analysis,

¹⁴While the baseline sample ends in 2022, I also obtain a supplemental update through 2024 and use it in Appendix [Figure A.10](#) to study the effect of the 2023–2024 high-rate environment.

I adopt the National Association of Insurance Commissioners (NAIC) system, which consolidates the three ratings into one and sorts them into six NAIC categories.¹⁵ The NAIC rating is the most relevant risk metric for insurance companies, as it determines the capital requirement for each bond. Bonds in NAIC 1 and NAIC 2 are investment-grade, while bonds in NAIC 3-6 are high-yield. Table 1 replicates Table 2 in Becker and Ivashina (2015), which summarizes the 5-year default rate and capital requirement of each NAIC category. Corporate bonds with lower NAIC ratings have higher default rates and entail more stringent capital requirements. Table A.1 contains additional summary statistics on bond yields, spreads, and other characteristics across different NAIC categories.

Table 1. The NAIC rating system

NAIC Category	Credit Ratings	Investment Grade	5-Year Default Rate (1990–2010)	Capital Requirement
NAIC 1 (highest)	AAA, AA, A	✓	0.00%, 0.09%, 0.69%	0.3%
NAIC 2	BBB	✓	2.62%	0.96%
NAIC 3	BB	x	6.76%	3.39%
NAIC 4	B	x	8.99%	7.38%
NAIC 5	CCC	x	34.38%	16.96%
NAIC 6 (lowest)	CC, C, D	x	n.a.	19.50%

This table summarizes the characteristics of corporate bonds belonging to different NAIC categories. The data on the cumulative 5-year default rates and capital requirements are drawn from Becker and Ivashina (2015).

Bond Indices and CDS For part of my empirical analysis, I use aggregate bond yield indices from the Intercontinental Exchange, which are accessed through FRED and began in 1997. I also use bond credit default swaps (CDS) spreads from Markit IHS. In particular, I use the par spreads of credit default swaps with a 5-year tenor, which form the most liquid segment of the CDS market (e.g., Blanco, Brennan, and Marsh 2005). The CDS data have been available since the beginning of 2001.

Life Insurers I obtain regulatory data on life insurers' end-of-year bond holdings and long-term bond transactions from NAIC between 2002 and 2019. I examine Schedule D

¹⁵The rating scheme only applies to publicly traded bonds. For private placement bonds not rated by credit rating agencies, ratings are assigned directly by the NAIC.

information in life insurers' regulatory reports to the NAIC. The NAIC data also contain bond identifiers (CUSIP codes) that allow me to match the bonds held and transacted by life insurers to those in Mergent FISD and WRDS Bond Returns. I use aggregate balance sheet data for life insurers from the Financial Accounts of the United States and life insurers' stock price data from the Center for Research in Security Prices.

FOMC Meeting Dates [Hillenbrand \(2025\)](#) summarized the dates of all FOMC meetings since September 1982. In my analysis, I focus on meetings between 1997 and 2022, excluding all unscheduled meetings.

Sample Period Throughout the paper, I separate my analysis into two time periods: before the Financial Crisis (pre-2007) and after the Financial Crisis (post-2009). I remove the Global Financial Crisis (2007–2009), which featured large-scale fire sales and liquidity disruptions that could confound the mechanism of interest. Similarly, I exclude data from March 2020 in the postcrisis period to avoid capturing the bond market disruptions during the COVID-19 crisis.

3 The Long-Term Interest Rate and Bond Credit Spreads

In this section, I examine the comovement between the long-term interest rate and corporate bond credit spreads and contrast the findings before and after the Financial Crisis.

3.1 Pass-through of the long-term interest rate

I begin by studying the 10-year Treasury yield pass-through in the cross section of corporate bonds. I run the following regressions to estimate the pass-through

$$\Delta y_{it} = \alpha_i + \sum_{k=1}^6 \beta_k \cdot \mathbf{1}_{\{\text{NAIC } k\}} \cdot \Delta y_t^{(10)} + \Gamma \mathbf{X}_{it} + \varepsilon_{it}, \quad (10)$$

where Δy_{it} is the change in bond i 's yield from month t to $t + 3$, $\Delta y_t^{(10)}$ is the change in the 10-year Treasury yield over the same period, α_i are bond fixed effects, and $\mathbf{X}_{it}$ are additional controls. Here, I control for bond characteristics, including bond

size (outstanding amount), liquidity (trading volume), time to maturity, credit ratings, coupon amount, coupon frequency, and the recent default rate for each NAIC category.

The regression coefficient β_k measures the pass-through rate of the 10-year Treasury yield to the yield of NAIC k corporate bonds. Intuitively, β_k measures the average response of the yields of NAIC k bonds when the 10-year Treasury yield increases by one percentage point. [Figure 2](#) shows the estimates of β_k for the sample before the Financial Crisis (pre-2007) and the sample after the Financial Crisis (post-2009).

After the Financial Crisis, there is a strong relationship between credit ratings and the pass-through of the long-term interest rate. The yields of the safest bonds (i.e., NAIC 1 and 2) show muted responses to the 10-year Treasury yield. However, as the credit rating declines, the comovement with the long-term interest rate becomes statistically significant and negative. The effects are large, especially for bonds in NAIC 4-6, whose yields decrease by much more than 1% when the 10-year Treasury yield increases by 1%. Accordingly, the credit spread between low- and high-rated bonds narrows when the 10-year Treasury yield is high. By contrast, the relationship was much weaker before the Financial Crisis. Bonds across all NAIC categories showed strong positive comovements with the 10-year Treasury yield. The responses of lower-rated bonds (NAIC 6) also appeared to be more positive than those of higher-rated bonds. Overall, the results indicate a significant change in the comovement between the long-term interest rate and corporate bond yields around the Financial Crisis.

3.2 Credit spread responses to the long-term interest rate

Next, I focus on credit spreads in the cross section of corporate bonds. To this end, I analyze the spreads between bonds with lower ratings (NAIC k , $k \geq 2$) and the highest credit rating (NAIC 1). Relative to the empirical specification in regression (10), I further include duration-time fixed effects $\alpha_{D(i),t}$,

$$\Delta y_{it} = \alpha_i + \alpha_{D(i),t} + \sum_{k=2}^6 \beta_k \cdot \mathbf{1}_{\{\text{NAIC } k\}} \cdot \Delta y_t^{(10)} + \Gamma \mathbf{X}_{it} + \varepsilon_{it}. \quad (11)$$

To create the duration-time fixed effects, I group bonds into small segments based on their Macaulay duration, each with a size of 1 year. I then interact these duration dummies with the observation date. The time fixed effects capture the yield movements of NAIC 1 bonds, and the coefficient β_k (for $k \geq 2$) measures the average response that

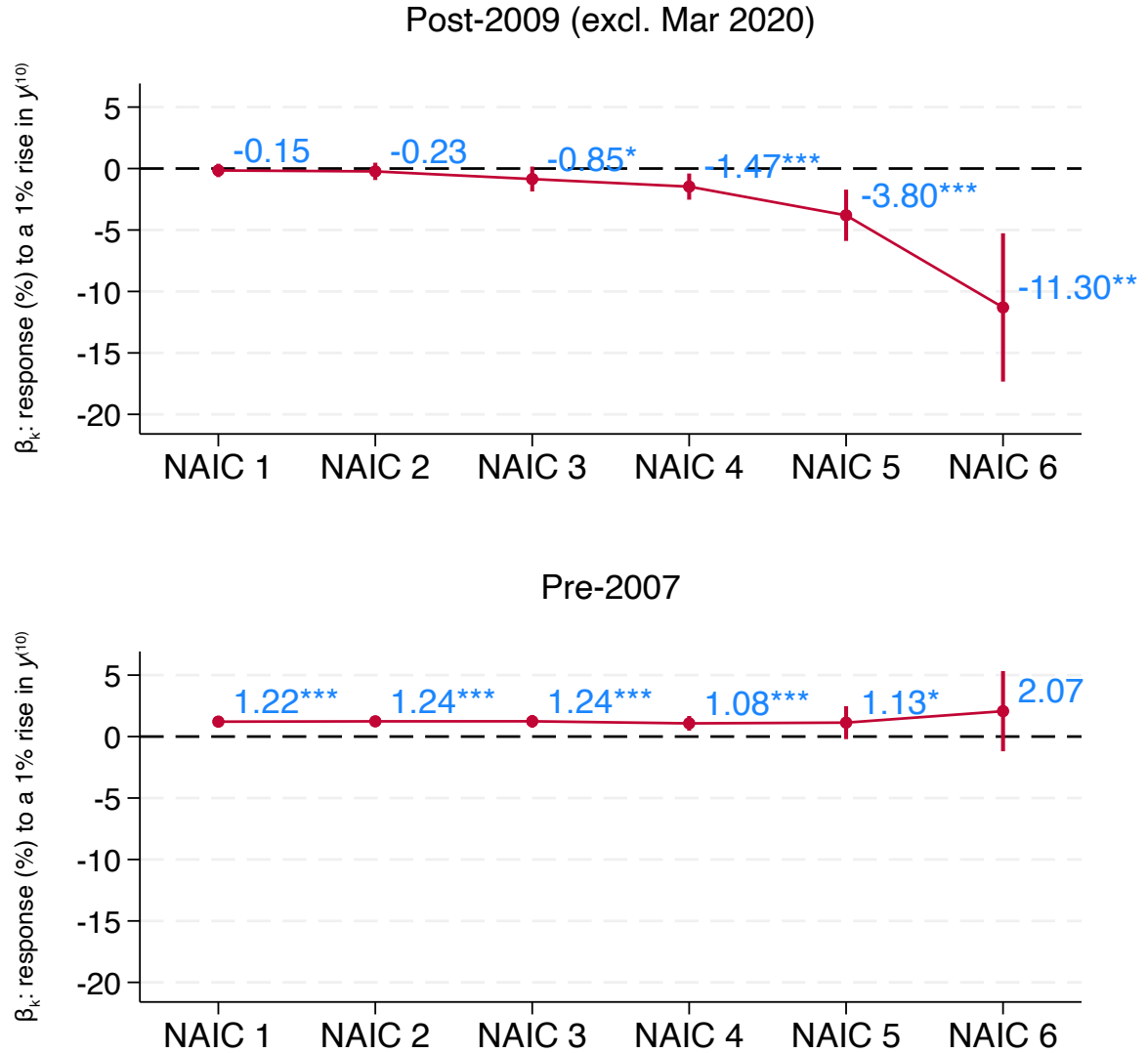


Figure 2. 10-year Treasury yield pass-through

This figure plots the coefficients β_k estimated from regression (10), controlling for the trading volume, outstanding amount, time to maturity, credit ratings, coupon rate, and coupon frequency of each bond, and the recent default rate of each NAIC category, all interacted with $\Delta y_t^{(10)}$. Both corporate bond yields and the Treasury yield are in percentage points. The top panel shows results for the postcrisis sample, while the bottom panel shows results for the precrisis sample. The statistical significance is based on standard errors clustered at the issuer and year-month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

changes in the 10-year Treasury yield induce in the spreads between NAIC k bonds and NAIC 1 bonds. The interaction of time fixed effects with duration fixed effects further ensures that I construct credit spreads using bonds of similar durations.¹⁶ A negative coefficient indicates that credit spreads move in a direction opposite that of the long-term interest rate.

Figure 3 plots the estimated coefficients and delivers one of the main results of this paper. In the postcrisis sample, corporate bond credit spreads narrow as the 10-year Treasury yield rises. The result is significant for all NAIC categories and is stronger for lower ratings.

The credit spread movements are large compared to the average credit spreads reported in Table A.1. When the 10-year Treasury yield increases by 1%, the credit spreads of NAIC 2 (BBB) bonds fall by 0.13%, which is equal to 15% of the average spread of NAIC 2 bonds. The effects are much larger for high-yield bonds. For example, when the 10-year Treasury yield increases by 1%, the yields of CCC corporate bonds (NAIC 5) fall by 3.69% relative to corporate bonds rated A or better (NAIC 1), which amounts to 1/3 of the average spread between NAIC 5 and NAIC 1. As already shown in Figure 2, the yields of high-yield bonds decline significantly as the 10-year Treasury yield rises. Surprisingly, increases in the long-term interest rate are associated with lower funding costs for the riskiest firms.

It is also worth noting that there were no effects on credit spreads before the Financial Crisis. In the sample before 2007, credit spreads showed very small, statistically insignificant responses to the 10-year Treasury yield. The earlier lack of response implies that the comovement between the long-term interest rate and credit spreads emerged only after the Crisis.

Figure A.1 and Figure A.2 in the Appendix visualize the time series of the 10-year Treasury yield and various corporate bond credit spreads. The correlation pattern supports the conclusion of this section. A strong negative correlation exists between corporate bond credit spreads and the 10-year Treasury yield after the Financial Crisis, but not before.

It is interesting to further examine whether the credit spread dynamics are driven by credit risk or other factors (such as insurers' risk-bearing capacity), as stronger economic

¹⁶As noted by van Binsbergen, Nozawa, and Schwert (2025), it is important to control for the duration when studying the pricing of corporate bonds. Also, riskier bonds tend to be of shorter duration, which makes their prices less sensitive to Treasury yields. Figure A.3 further confirms that the results also hold when we compare bonds within duration segments.

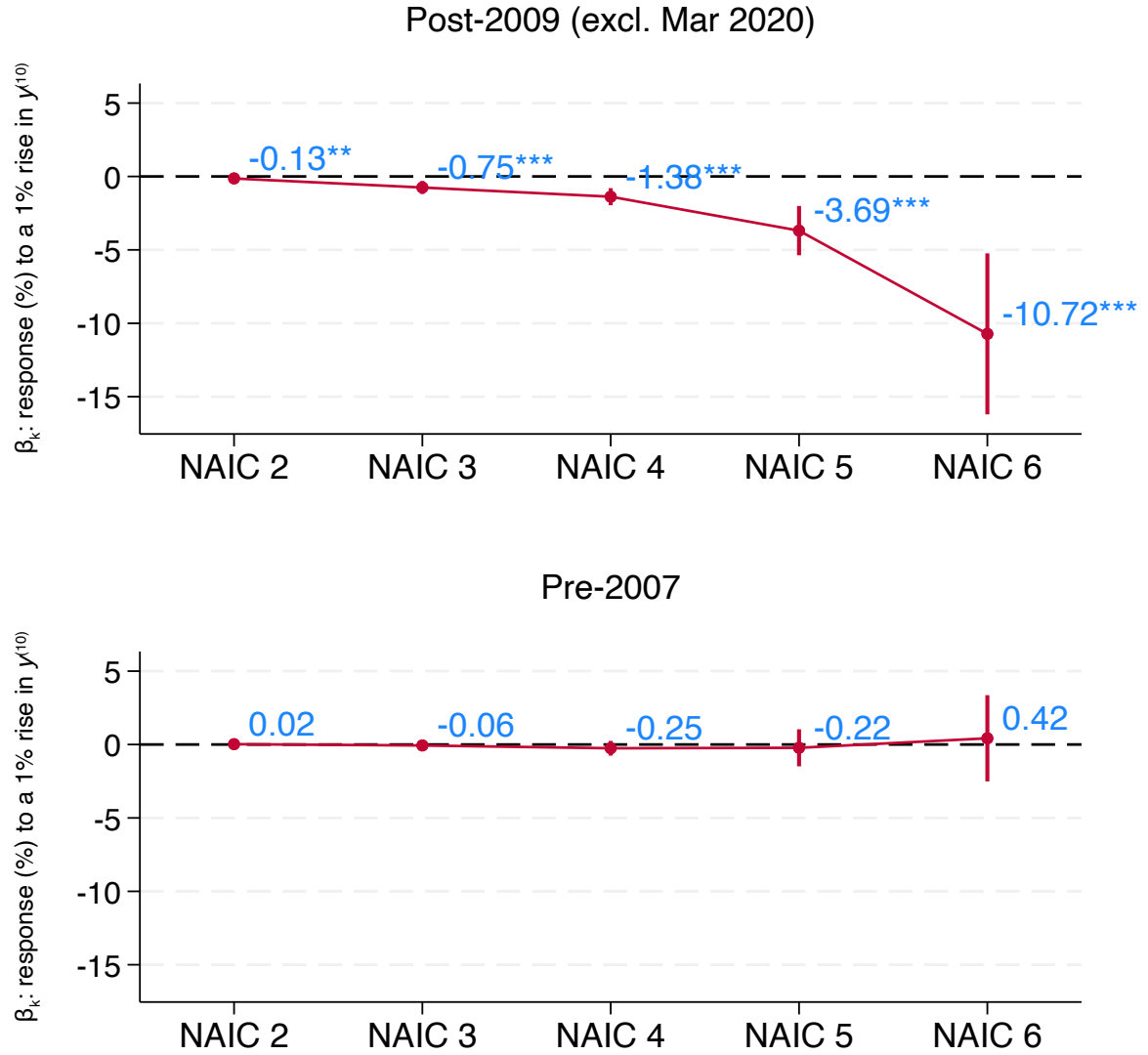


Figure 3. Credit spread responses to the long-term interest rate

This figure plots the coefficients β_k estimated from regression (11), controlling for the trading volume, outstanding amount, time to maturity, credit ratings, coupon rate, and coupon frequency of each bond, and the recent default rate of each NAIC category, all interacted with $\Delta y_t^{(10)}$. Both corporate bond yields and the Treasury yield are in percentage points. The top panel shows results for the postcrisis sample, while the bottom panel shows results for the precrisis sample. The statistical significance is based on standard errors clustered at the issuer and year-month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

growth can reduce firms' credit risk, which could further lead to both lower credit spreads and increased interest rates (e.g., [Wu and Zhang 2008](#)). To account for variations in credit risk, I control for changes in bond issuers' EDF based on the [Merton \(1974\)](#) distance to default model following [Bharath and Shumway \(2008\)](#).¹⁷ Column 4 in Internet Appendix Table F.1 shows the coefficients estimated in the sample of publicly traded firms after controlling for EDF, which exhibit the same pattern as in [Figure 3](#) and confirm that credit risk itself has little effect on the negative comovements between long rates and credit spreads.

Another important potential confounding factor is the value of call options. As noted in [Ma, Streitz, and Tourre \(2023\)](#), most corporate bonds issued in recent periods have embedded call options. Typically, bond prices fall when interest rates rise, reducing the likelihood that call options will be exercised. Hence, call options can also contribute to negative comovements between long rates and credit spreads if bonds with different credit ratings differ systematically in the moneyness of their call options ([Longstaff and Schwartz 1995](#); [Duffee 1998](#)). To understand whether the findings in [Figure 3](#) are mostly driven by call options, I collect the call prices and moneyness ratio (bond price/call price) of a large fraction of bonds in my sample.¹⁸ Column 3 in Internet Appendix Table F.1 shows that the results remain largely the same after controlling for callability type dummies and the call option moneyness interacted with $\Delta y_t^{(10)}$ at the bond level. The results show that the value of call options can at most explain a small fraction of the dynamics of interest rate credit spreads.

3.3 High-frequency evidence: FOMC announcements

Sections 3.1 and 3.2 discussed the unconditional comovements between the long-term interest rate and corporate bond credit spreads. In this section, I strengthen the evidence by showing that the estimated relationship between 10-year Treasury yields and bond credit spreads is robust to high-frequency shocks around FOMC meetings. A recent study by [Hillenbrand \(2025\)](#) shows that a short window around FOMC meetings accounts for most of the long-run movements in long-term interest rates over the past 30 years. A potential explanation is that FOMC meetings disseminate information about

¹⁷[Bharath and Shumway \(2008\)](#) propose a way of constructing Merton EDF for public firms using Compustat data. Using their method and the implementation by [Gao \(2022\)](#), I can compute the Merton EDF for 75.64% of observations in my data set.

¹⁸I am able to recover the moneyness for 66.44% of observations in my data set. The procedure of constructing call prices is detailed in [Appendix B](#).

future interest rate paths. Inspired by [Hillenbrand \(2025\)](#), I construct shocks to the 10-year Treasury yield as the changes in the yield in 2-day windows around FOMC meetings,

$$\Delta y_t^{(10)} \Big|_{\text{FOMC}} = y_{t+1}^{(10)} - y_{t-1}^{(10)},$$

where t is an FOMC announcement day. Using a local projection method ([Jordà 2005](#)), I then estimate the impulse responses of corporate bond credit spreads to the high-frequency 10-year Treasury yield shocks. The regressions are as follows

$$Spread_{t+h}^k - Spread_{t-1}^k = \alpha_h + \beta_h \left(\Delta y_t^{(10)} \Big|_{\text{FOMC}} \right) + \varepsilon_{t,h}. \quad (12)$$

Here, the left-hand-side variable $Spread_t^k$ is constructed as the difference between the option-adjusted spread (OAS) indices for rating k and the AAA rating,

$$Spread_t^k = OAS_t^k - OAS_t^{AAA}.$$

The index OAS_t^k is constructed as the weighted OASs of all bonds in rating k measured against the spot Treasury yield curve. The coefficient β_h then measures the cumulative response of $Spread_t^k$ to a 1% positive innovation in the 10-year Treasury yield at a horizon of h trading days.

Figures 4 and 5 plot the impulse response functions estimated from the local projections (12) for the postcrisis sample and the precrisis sample, respectively. After the Financial Crisis, positive shocks to the 10-year Treasury yield led to large and significant declines in the credit spreads between AAA bonds and lower-rated bonds. For example, a 1% increase in the 10-year Treasury yield depresses the spread between single B and AAA bonds around 1% at a 10-day horizon. The negative responses of credit spreads are surprising in light of the literature on short-term monetary policy, which typically finds that higher interest rates raise risk premia and corporate bond spreads (e.g., [Gertler and Karadi 2015](#)).

Nevertheless, similar shocks to the 10-year Treasury yield produced statistically insignificant, or even opposite, responses before the Crisis. The results suggest that there was a shift in the transmission of policy shocks around the Financial Crisis—the FOMC windows did not influence credit spreads prior to the Financial Crisis.

[Figure A.5](#) estimates the local projections directly on the yield indices for different

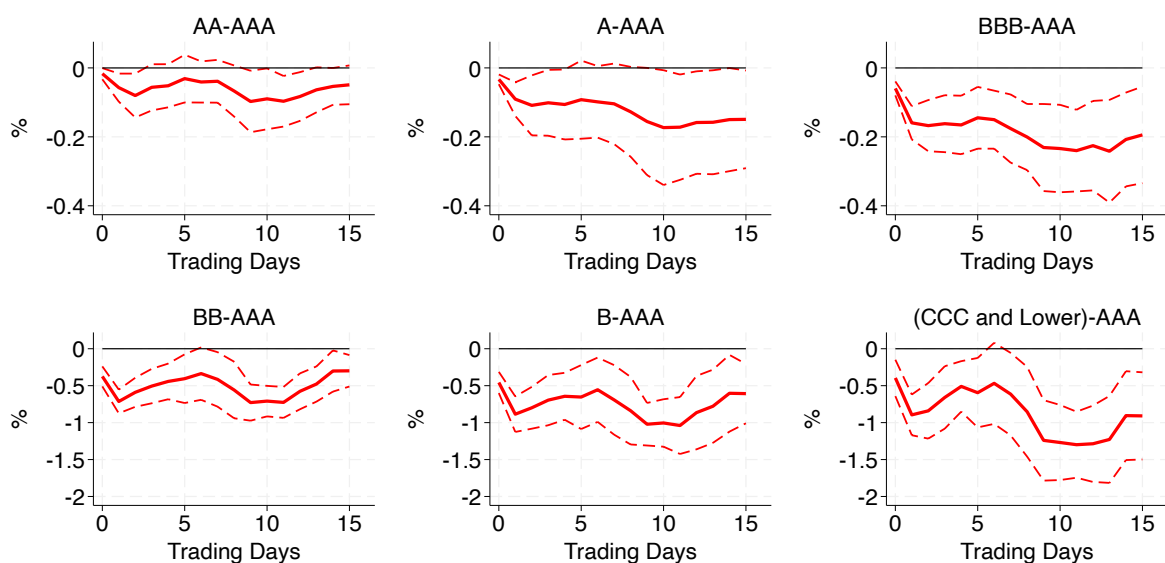


Figure 4. Impulse responses of credit spreads (2010–2022)

This figure plots the cumulative responses of credit spreads to a 1% increase in the 10-year Treasury yield. The dashed lines show 90% confidence intervals based on Newey-West standard errors estimated with a maximum lag of 30 trading days.

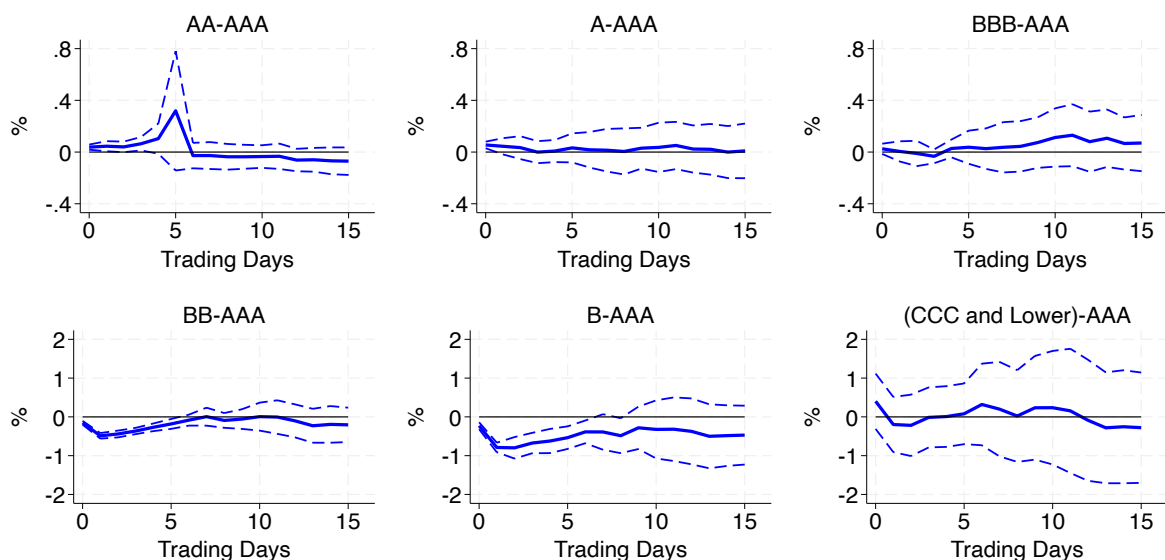


Figure 5. Impulse responses of credit spreads (1997–2007)

This figure plots the cumulative responses of credit spreads to a 1% increase in the 10-year Treasury yield. The dashed lines show 90% confidence intervals based on Newey-West standard errors estimated with a maximum lag of 30 trading days.

credit ratings. The yield of AAA bonds exhibits a persistent positive relationship with the 10-year Treasury yield, whereas yields of other investment-grade bonds (AA, A, BBB)

are less sensitive. Consistent with the findings of [Section 3.1](#), the yields of high-yield bonds (BB and lower) move in the direction opposite that of the 10-year Treasury yield.

An alternative explanation is that the effects are mainly due to short rates moving in step with long rates. As a robustness check, I include changes in the 1-month Treasury yield around FOMC meetings as controls for short-term monetary policy. Further, long-rate movements around FOMC meetings could contain information on credit risk and the long-run performance of corporate bonds. To control for shocks to credit risk, I further control for the changes in the average CDS spread for each NAIC category against AAA bonds around FOMC meetings, which is a direct and real-time measure of the price of default risk (e.g., [Longstaff, Mithal, and Neis 2005](#)).¹⁹ By controlling for CDS spreads, we can eliminate the effects of changing credit risk and more clearly understand how long rates affect credit spreads.²⁰ [Figure A.4](#) plots the postcrisis impulse responses estimated after controlling for short rate changes and CDS spreads, which again confirm the patterns shown in [Figure 4](#).

Another potential confounding mechanism is the “Fed response to news” channel, where long rate increases around FOMC meetings reflect policy responses to improved economic conditions. However, while the “Fed response to news” channel is important for understanding short rate surprises around FOMC meetings ([Bauer and Swanson 2023](#)), there is no evidence that it plays a significant role in the dynamics of long rates. [Hillenbrand \(2025\)](#) finds that changes in economic conditions before FOMC meetings do not predict changes in the 10-year Treasury yield around FOMC meetings. [Gürkaynak, Sack, and Swanson \(2005\)](#) show that long rates actually tend to fall in response to FOMC tightenings, which are more likely to occur after improved economic conditions. Indeed, [Bauer and Swanson \(2023\)](#) also argued that “an information effect is not needed to explain the response of long-term Treasury yields to FOMC announcements.”

3.4 The long-term interest rate and bond issuance

In previous sections, I documented that decreases in long-term interest rates are accompanied by increases in credit spreads, especially for high-yield bonds. Credit

¹⁹Absent frictions, CDS spreads should coincide with credit spreads. However, empirically, the CDS-bond basis is known to be nonzero and dispersed across bonds (e.g., [Bai and Collin-Dufresne 2019](#)). Following [Longstaff, Mithal, and Neis \(2005\)](#) and [Mota \(2023\)](#), I view the CDS spread as a better measure of the market price of credit risk, whereas credit spreads may also reflect other factors, including liquidity, convenience yields, and the risk-bearing capacity of bond investors, such as life insurers.

²⁰Importantly, life insurers are not the major investors in the CDS market, so the CDS spreads are potentially not affected by the balance sheets of life insurers.

spreads are important indicators of firm borrowing costs—existing studies show that primary market bond prices (i.e., new bond issuance costs) are highly correlated with secondary market conditions (e.g., [Coppola 2025](#); [Flanagan, Kedia, and Zhou 2019](#)). Therefore, the comovement between the long-term interest rate and corporate bond credit spreads likely affects firms’ incentives to issue new bonds.

To investigate the comovement between the long-term interest rate and corporate bond issuance, I aggregate new bond issuance for investment-grade (IG, NAIC 1-2) and high-yield (HY, NAIC 3-6) bonds in the WRDS Bond Returns data set. I then estimate the following empirical specification

$$\Delta \text{Issuance Rate}_{k,t} = \alpha_t + \alpha_k + \beta \cdot \mathbf{1}_{\{k=\text{HY}\}} \cdot \Delta y_t^{(10)} + \Gamma \mathbf{X}_{kt} + \varepsilon_{kt}, \quad (13)$$

where $k \in \{\text{IG}, \text{HY}\}$, $\text{Issuance Rate}_{k,t}$ is defined as the total IG/HY issuance in the 6-month period after month t divided by the current total outstanding amount of IG/HY bonds at the end of the month t , $\Delta y_t^{(10)}$ is the change in the 10-year Treasury yield from month t to $t + 3$, $\Delta \text{Issuance Rate}_{k,t}$ is the change in the issuance rate over the same period, and the controls $\mathbf{X}_{kt}$ include the recent default rates of IG and HY bonds. The rating fixed effects α_k capture the steady state difference in the growth rates of IG and HY markets. The time fixed effects α_t capture aggregate fluctuations in bond issuance. The coefficient β measures how the 10-year Treasury yield affects the difference in issuance between IG and HY bonds. The issuance rate of HY bonds increases by β relative to IG bonds when the 10-year Treasury yield increases by 1%.

[Table 2](#) contains the estimated coefficients from regression (13). The second column reports the results after the Financial Crisis. It suggests that increases in the 10-year Treasury yield boost the issuance of high-yield bonds relative to investment-grade bonds. Following a 1% increase in the long-term interest rate, the issuance of high-yield (NAIC 3-6) bonds grows by more than 0.9% relative to the issuance of investment-grade (NAIC 1-2) bonds. The results suggest that the long-term interest rate potentially alters the composition of the corporate bond market. Investment-grade bonds account for a larger share of total new issuance when long-term interest rates are low. In contrast, the issuance differences are not correlated with the long-term interest rate before the Financial Crisis. The lack of correlation is expected since the long-term interest rate affected all corporate bonds equally during that period, as demonstrated in [Section 3.1](#).

After the Financial Crisis, the U.S. economy entered a sustained period of low interest rates, accompanied by a large expansion of the bond market. Consistent with

Table 2. Bond issuance responses to the long-term interest rate

	Pre-2007	Post-2009
$\mathbf{1}_{\{k=HY\}} \times \Delta y_t^{(10)}$	-0.136 [-0.371]	0.939** [2.006]
NAIC FE	✓	✓
Time FE	✓	✓
R^2	.589	.725

This table shows the coefficients estimated from regression (13), controlling for the recent default rates of each NAIC category and the average issuance maturity interacted with $\Delta y_t^{(10)}$. The first column shows results for the precrisis sample, while the second column shows results for the postcrisis sample. Both the response variable and the 10-year Treasury yield are in percentage points. The t -statistics shown in brackets are based on standard errors clustered at the IG/HY category by year-month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

my findings, [Figure 6](#) shows that the corporate bond market has shifted towards the investment-grade segment during the low-interest-rate environment post-2008 (see also [Mota 2023](#)).

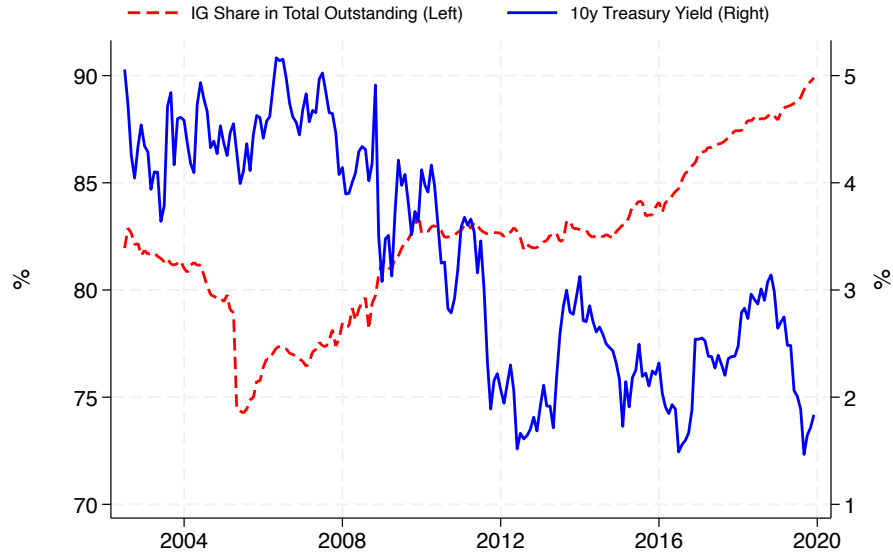


Figure 6. Share of investment-grade bonds in WRDS bond returns

4 The Role of Life Insurers

Next, I turn to the potential channel through which the long-term interest rate negatively affects corporate bond credit spreads—life insurers’ duration mismatch. As shown in [Figure 1](#), life insurers are the largest investor group in corporate bonds and own more than 20% of all corporate bonds in recent years ([Kojen and Yogo 2023](#)).

4.1 Duration mismatch

Modern life insurers issue various life insurance and annuity products while investing primarily in fixed-income markets. For example, variable annuities, long-term mutual fund products with minimum-return guarantees, have now become the largest component of life insurers’ liabilities ([Kojen and Yogo 2022](#)). While insurers manage their portfolios with asset-liability considerations in mind, their assets and liabilities need not be perfectly duration matched due to various financial frictions. In particular, life insurers’ long-duration liabilities can be more sensitive to interest rates than their fixed-income assets, especially when interest rates are low. Hence, life insurers’ equity could be hurt when long-term interest rates fall.

[Table 3](#) summarizes the balance sheet structure of the U.S. life insurance sector. Life insurers’ assets are predominantly fixed-income debt securities and mutual fund shares that are ultimately also invested in fixed-income assets. Corporate bonds are an important asset class for life insurers as they make up more than 70% of all the debt securities they hold. On the liability side, life insurers raise funding primarily by selling life insurance and annuities, which typically have longer maturities and durations than corporate bonds.

I then estimate the exposure of life insurers’ market equity to the 10-year Treasury yield. In particular, I estimate the following regression

$$InsurerReturn_t = \alpha + \beta \Delta y_t^{(10)} + \gamma MarketReturn_t + Controls + \varepsilon_t. \quad (14)$$

$InsurerReturn_t$ is the stock return of the life insurance sector in week t , $MarketReturn_t$ is the S&P 500 return in week t , and $\Delta y_t^{(10)}$ is the change in the 10-year Treasury yield in week t . The coefficient β measures the sensitivity of life insurers’ market equity to changes in the 10-year Treasury yield. If the duration of life insurers’ liabilities exceeds their assets, the coefficient β should be positive, as higher interest rates lower the value

Table 3. The balance sheets of the life insurance sector

Financial Assets (\$ tn)		Liabilities (\$ tn)	
Short-term Assets	0.20	Life Insurance Reserves	2.25
Debt Securities	4.43	Annuity Reserves	3.79
–Corporate Bonds	–3.56	Other Liabilities	2.34
Loans & Equities	1.54		
Mutual Fund Shares	1.70		
Other Financial Assets	1.55		
Total	9.42	Total	8.38

This table summarizes the main financial assets and liabilities of life insurers in the United States. The data is from the Financial Accounts of the United States at 2020 Q4.

of their liabilities more than their assets and thereby boost the value of their equity. Therefore, $(-\beta)$ can also be interpreted as an estimate of the duration of life insurers' market equity, which is defined as $D_E = -\partial E_t / \partial y_t^{(10)}$ (E_t is the market equity of life insurers in week t). A positive estimate of β then indicates that life insurers' equity has a negative duration (e.g., [Ozdagli and Wang 2019](#); [Koijen and Yogo 2022](#)).

[Table 4](#) contrasts life insurers' exposure to the 10-year Treasury yield before and after the Financial Crisis. Before the Financial Crisis, the stock returns of the life insurance sector did not appear to be affected by long-term interest rates. It implies that life insurers were largely hedged against interest rate fluctuations. However, the estimated β is significantly far from zero in the postcrisis period. The estimated coefficient of 6.369 suggests that life insurers' equity value increases by more than 6% when the 10-year rate rises by 1%, indicating that life insurers faced a severe duration mismatch after the Financial Crisis.²¹

[Figure 7](#) plots the market leverage ratio of the life insurance sector against the 10-year Treasury yield. Consistent with the previous findings, life insurers become less levered when the long rate is high, as their liabilities become less expensive and their equity increases.

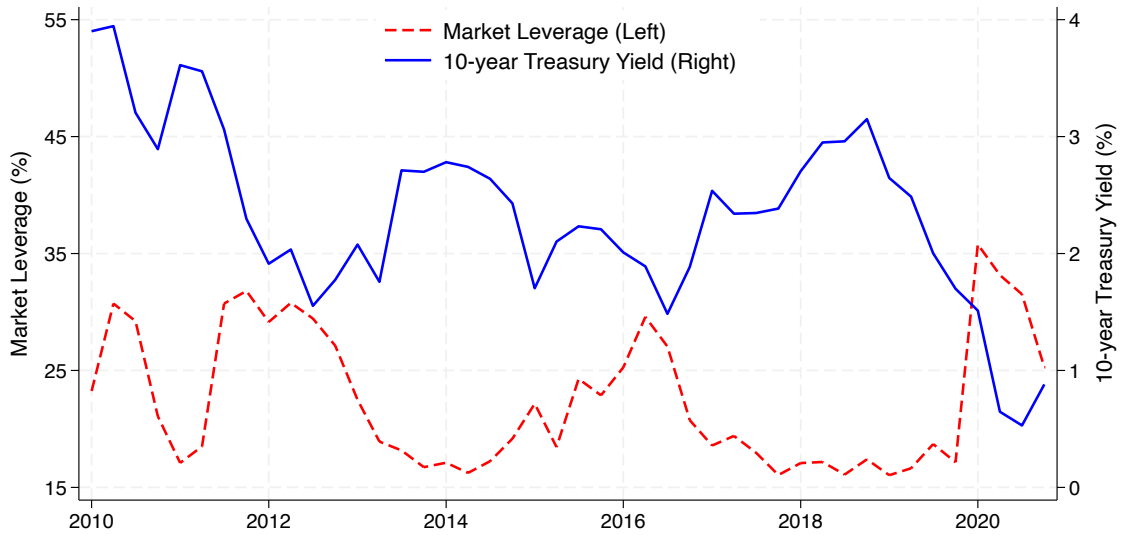
The Source of Duration Mismatch In this paper, I take a neutral stance on the cause of the shift in life insurers' duration mismatch. Nevertheless, the existing literature

²¹In Internet Appendix Table F.3, I find that insurers' duration mismatch has greatly reduced during the recent high-interest-rate period of 2023–2024. Meanwhile, [Figure A.10](#) shows that the negative comovements between long rates and credit spreads have also diminished.

Table 4. Life insurers' duration mismatch

	Pre-2007	Post-2009
$\Delta y_t^{(10)}$	-0.009 [-0.008]	6.369*** [6.665]
S&P 500 Return	1.037*** [14.457]	1.192*** [18.706]
R^2	.648	.728
Observations	260	676

This table shows the coefficients estimated from regression (14) between 2002 and 2022, controlling for the market return. The first column shows the result for the precrisis sample, while the second column shows the result for the postcrisis sample. Both the excess return and the 10-year Treasury yield are in percentage points. The t -statistics shown in brackets are based on robust standard errors. * $p < .1$; ** $p < .05$; *** $p < .01$.

**Figure 7.** Life insurers' market leverage

This figure plots the market leverage ratio of the U.S. life insurance sector and the 10-year Treasury yield. The market leverage ratio is defined as the “fair” value of liabilities (from Compustat) divided by market equity value.

typically attributes the new duration mismatch post-2008 to lower interest rates after the Financial Crisis. To illustrate this point, we can write life insurers' equity as $E = A - L$, where A is assets, L is liabilities, and $\ell = L/E$ is the leverage ratio. The duration of equity is then $D_E = (AD_A - LD_L)/E = (1 + \ell)D_A - \ell D_L$. Before the Crisis, life insurers were hedged against interest rate risk by choosing $D_A \approx [\ell/(1 + \ell)]D_L$ so that $D_E \approx 0$.

Meanwhile, the assets and the liabilities both have a positive convexity as lower interest rates increase bond duration,

$$C_A := -\frac{\partial D_A}{\partial y_t^{(10)}} > 0, \quad C_L := -\frac{\partial D_L}{\partial y_t^{(10)}} > 0.$$

Crucially, the convexity of life insurers' liabilities is greater than the convexity of their assets ($C_L > C_A$), meaning that D_L rises faster than D_A when interest rates fall.²² As a result, D_E became negative in the low-interest-rate environment after the Crisis.

Furthermore, there exist other market and institutional frictions that prevent life insurers from increasing their asset duration D_A to close the duration gap. Domanski, Shin, and Sushko (2017) and Greenwood and Vissing-Jorgensen (2018) argue that large-scale portfolio rebalancing of life insurers towards longer-term assets puts downward pressure on the long-term interest rate, which could further exacerbate the duration mismatch. Ozdagli and Wang (2019) emphasize the role of portfolio adjustment frictions resulting from the cost of large bond trades. Kojen and Yogo (2021) discussed several other reasons why life insurers do not fully hedge their interest rate exposure, including market incompleteness (i.e., the scarcity of long-term assets and options), risk-shifting motives, and regulatory distortions. In particular, Huber (2022) and Sen (2023) argue that the regulatory framework imposed on life insurers might not properly capture the interest rate risk and thus distorts life insurers' hedging incentives.²³

4.2 Life insurers and credit spread dynamics

I further show that life insurers play an important role in shaping the comovement between the long-term interest rate and corporate bond credit spreads. Section 3.2 illustrates that long-term interest rates have heterogeneous impacts on the cross section of corporate bonds after the Financial Crisis. If life insurers' duration mismatch is the main channel through which the long-term interest rate affects credit spreads, we should expect the pattern to be more pronounced in bonds held by life insurers, as confirmed by the following results.

²²One likely explanation is that life insurers' liabilities have a longer maturity than their assets, as the convexity of an asset typically increases in its maturity. It could also be because the embedded options in some of life insurers' variable annuity liabilities are less exercised when interest rates are low (e.g., Ozdagli and Wang 2019; Kojen and Yogo 2022).

²³For more examples of regulatory distortions in the insurance sector, see also Lee, Mayers, and Smith (1997), Ellul, Jotikasthira, and Lundblad (2011), Becker, Opp, and Saidi (2022), and Ellul et al. (2022).

First, I provide suggestive evidence that life insurers matter for credit spread dynamics by estimating regression (11) in Section 3.2 separately for two subsamples: bonds that life insurance companies hold and bonds that life insurance companies do not hold. Figure A.8 plots the estimated coefficients from the two separate regressions for the postcrisis period. Indeed, the negative relationship between credit spreads and long rates in Figure 3 is only present in bonds with positive life insurance ownership.

To further establish the link between life insurers' duration mismatch and credit spread dynamics, I augment regression (11) with an interaction term between insurers' risk exposure and $\Delta y_t^{(10)}$,

$$\Delta y_{it} = \alpha_i + \alpha_{D(i),t} + \sum_{k=2}^6 (\beta_k + \gamma_k \text{Exposure}_t) \cdot \mathbf{1}_{\{\text{NAIC } k\}} \cdot \Delta y_t^{(10)} + \Gamma \mathbf{X}_{it} + \varepsilon_{it}, \quad (15)$$

where Exposure_t is insurer's exposure to long rates estimated from (14) using data 1 year prior to date t . If we estimate $\gamma_k < 0$, we know that the comovements between long rates and credit spreads become more negative when insurers have more negative duration gaps. Figure A.9 summarizes the γ coefficients estimated from regression (15), which again confirms that the comovements between long rates and credit spreads are negatively linked to the size of the life insurance sector's duration gap. Moreover, the negative association is stronger for lower credit ratings, which is consistent with the findings of Section 3.2.

Next, I recover the fraction of each bond owned by life insurers from NAIC regulatory reports and study the role of life insurance ownership in shaping the results of Section 3.2.²⁴ To further sharpen the identification, I utilize an exogenous discontinuity in investor composition stemming from mutual funds' investment mandates (e.g., Li and Yu 2026; Bretscher, Schmid, and Ye 2023). Bai, Li, and Manela (2023) and Li and Yu (2026) document that a large fraction of corporate bond funds are "intermediate-term" and are mandated to only invest in bonds whose maturity is less than 10 years. A bond with a maturity greater than 10 years at issuance will experience a surge in demand from mutual funds once it ages to the point where its maturity drops below 10 years. As a result, the bond's ownership by other investors, such as life insurance companies, jumps downward.

Figure 8 visualizes the discontinuity in life insurers' ownership share around the

²⁴In the main regression of this section, I use end-of-year observations as insurance companies' holdings data is only recorded at the end of each year.

maturity threshold of 10 years and a quarter.²⁵ Indeed, we observe a large discontinuous jump in life insurers' ownership shares at the maturity threshold. Bonds whose maturity falls below the threshold are much less likely to be held by life insurers. Figures A.6 and A.7 show that the discontinuity disappears if we use other cutoff values such as 9 and 11 years.

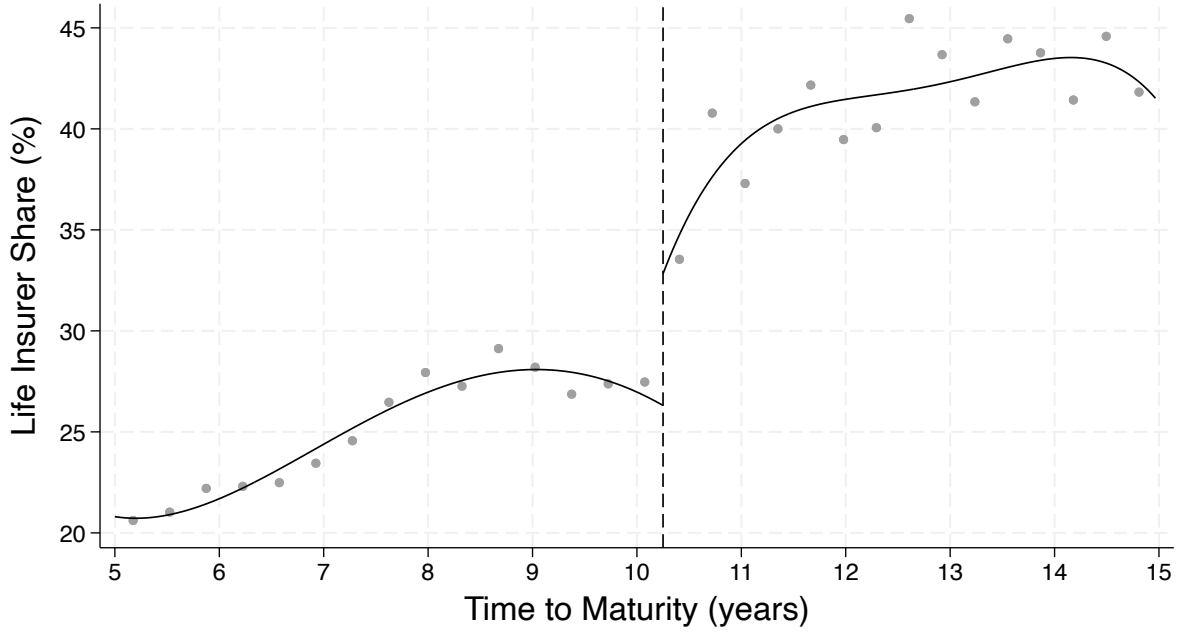


Figure 8. Discontinuity in investor composition

This figure shows a bin scatter plot of corporate bonds' life insurance ownership share (%) and their time to maturity. The vertical line indicates the maturity threshold of 10.25 years. The figure also shows the best-fit fourth-order polynomials for observations on each side of the threshold.

I further test the existence of the discontinuity using both OLS regressions and the robust bias-correction method of [Calonico, Cattaneo, and Titiunik \(2014\)](#). In the OLS method, I regress the life insurer ownership share on a dummy variable indicating whether a bond's maturity is above the threshold $\mathbf{1}_{\{\text{maturity}_{it} \geq c\}}$ where $c = 10.25$,

$$\varphi_{it}^{\text{Ins}} = \alpha + \delta \cdot \mathbf{1}_{\{\text{maturity}_{it} > c\}} + \Gamma \mathbf{X}_{it} + \varepsilon_{it}.$$

²⁵The actual maturity cutoff for the discontinuity (10.25 years) is slightly more than 10 years. A likely explanation is mutual funds' window-dressing behavior, meaning that mutual funds only need to comply with their mandates at the end of each quarter when they disclose their bond holdings ([Morey and O'Neal 2006](#); [Agarwal, Gay, and Ling 2014](#)). Therefore, mutual funds could invest in bonds with a maturity between 10 and 10.25 years, even if they are mandated to invest in bonds with a maturity of less than 10 years.

In the robust bias-correction method, I treat the problem as a sharp regression discontinuity design (RDD), where the treatment status is determined by $\mathbf{1}_{\{\text{maturity}_{it} \geq c\}}$ and the outcome variable is the life insurer ownership share.

Table 5 summarizes the effect of the maturity threshold on life insurer shares estimated using different methods. All results imply a strong discontinuity in investor composition at the maturity cutoff. The life insurers' ownership share is about 4–5% higher for bonds with maturity slightly above the cutoff.

Table 5. Testing the discontinuity

Method	δ	p -value	[95% Conf. Interval]
OLS	4.73	0.000	[4.15, 5.32]
RDD, Conventional	4.43	0.000	[2.46, 6.39]
RDD, Bias-Corrected	4.52	0.000	[2.55, 6.48]
RDD, Bias-Corrected, Robust	4.52	0.000	[2.22, 6.81]

This table shows the coefficient on $\mathbf{1}_{\{\text{maturity}_{it} > c\}}$, controlling for the trading volume, duration, maturity, size, coupon rate, and coupon frequency for each bond, and the recent default rate in each NAIC category. The first row shows the OLS estimate with robust standard errors, while the other rows show the estimates from various RDD methods discussed in [Calonico, Cattaneo, and Titiunik \(2014\)](#).

I then examine the effect of life insurance ownership on the response of a bond's credit spread to the long-term interest rate. I use the fuzzy RDD approach by instrumenting the life insurer share $\varphi_{it}^{\text{Ins}}$ with the dummy variable $\mathbf{1}_{\{\text{maturity}_{it} \geq c\}}$ for observations near the threshold (e.g., [Lee and Lemieux 2010](#)). The identification relies on the discontinuity of the investor base around the threshold. In other words, I compare the comovement seen in bonds slightly above the threshold to that seen in bonds slightly below the threshold. Specifically, I estimate the following two-stage regression:

$$\begin{aligned} \varphi_{it}^{\text{Ins}} &= \alpha + \delta \cdot \mathbf{1}_{\{\text{maturity}_{it} > c\}} + \Gamma \mathbf{X}_{it} + \varepsilon_{it} \\ \Delta y_{it} &= \alpha_t + \left(\beta + \gamma \varphi_{it}^{\text{Ins}} \right) \mathbf{1}_{\{\text{NAIC } 3-6\}} \Delta y_t^{(10)} + \Gamma \mathbf{X}_{it} + \varepsilon_{it}. \end{aligned} \quad (16)$$

The first stage estimates the effect of the maturity cutoff on the life insurer share $\varphi_{it}^{\text{Ins}}$. In the second stage, I use the optimal bandwidth derived in the first stage (2.512 years pre-GFC and 1.858 years post-GFC), following the method of [Calonico, Cattaneo, and Titiunik \(2014\)](#).²⁶

²⁶Table A.4 shows the estimates using alternative bandwidths around the optimal bandwidth.

The second stage estimates the effect of the life insurer share $\varphi_{it}^{\text{Ins}}$ on the comovement between the long-term interest rate and credit spreads, using $\mathbf{1}_{\{\text{maturity}_{it} > c\}}$ as an instrumental variable.²⁷ In the second stage, I focus on the credit spread between investment-grade bonds (NAIC 1-2) and high-yield bonds (NAIC 3-6). The time fixed effects α_t absorb the average responses of IG bond yields to the 10-year Treasury yield. The term $(\beta + \gamma \varphi_{it}^{\text{Ins}}) \mathbf{1}_{\{\text{NAIC } 3-6\}} \Delta y_t^{(10)}$ captures the responses of IG-HY credit spreads to the 10-year Treasury yield. Further, we can decompose the credit spread responses into two components: the first term $\beta \cdot \mathbf{1}_{\{\text{NAIC } 3-6\}} \Delta y_t^{(10)}$ captures the responses of credit spreads of bonds not held by life insurance companies, and the interaction term $\gamma \cdot \varphi_{it}^{\text{Ins}} \mathbf{1}_{\{\text{NAIC } 3-6\}} \Delta y_t^{(10)}$ captures the degree to which the credit spread responses depend on the bonds' life insurance ownership.

Table 6 presents the estimates from the RDD regressions above. The first-stage F -stats, both before and after the Financial Crisis, are well above the conventional threshold for strong instruments in Stock and Yogo (2005), again confirming the validity of the discontinuity in the bond ownership structure.

Table 6. RDD regressions

	Pre-2007	Post-2009
γ	-0.099 [-0.059]	-9.038** [-2.675]
Time FE	✓	✓
Controls	✓	✓
First-stage F	54.467	110.164
Observations	4,506	7,894

This table shows the coefficients estimated from regression (16), with the instrument, controlling for the trading volume, time to maturity, size, coupon rate and frequency, credit rating for each bond, and the recent default rate for each NAIC category, all interacted with $\Delta y_t^{(10)}$, and the bonds' insurer ownership share. The insurance ownership is normalized by its standard deviation. The first column shows the result for the precrisis sample, while the second column shows the result for the postcrisis sample. The t -statistics shown in brackets are based on standard errors clustered at the issuer and year-month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

After the Financial Crisis, we obtained negative and significant coefficients on $\varphi_{it}^{\text{Ins}} y_t^{(10)}$. The results show that life insurer holdings amplify the negative comovement between the long-term interest rate and credit spreads. Consider two high-yield bonds, with the second's life insurance ownership one standard deviation higher. The

²⁷I instrument $\varphi_{it}^{\text{Ins}} \mathbf{1}_{\{\text{NAIC } 3-6\}} \Delta y_t^{(10)}$ with $\mathbf{1}_{\{\text{maturity}_{it} > c\}} \mathbf{1}_{\{\text{NAIC } 3-6\}} \Delta y_t^{(10)}$.

estimates suggest that the credit spread of the second bond would decline by 9.038% more than the first when the 10-year rate increases by 1%. It is worth noting that the special role of life insurers began after the Financial Crisis, as the coefficient on $\varphi_{it}^{\text{Ins}} y_t^{(10)}$ is statistically insignificant before the GFC.²⁸

A threat to identification is that bond issuers might try to cater to mutual fund investors by issuing bonds just below 10 years, so the bonds on the two sides of the cutoff are systematically different. To alleviate such concerns, I conduct two additional robustness checks. The first exercise excludes recently issued bonds, and the second focuses on bonds with maturities above the maturity cutoff at issuance. The first two columns of [Table A.5](#) show that the results survive these robustness checks.²⁹

4.3 Bond transactions

So far, I have shown that declines in the long-term interest rate have a greater effect on bonds held by life insurers after the Financial Crisis. The main hypothesis is that life insurers' risk-bearing capacity becomes more restricted when lower long-term interest rates erode their equity (see [Section 4.1](#)). To further verify the mechanism, I examine the bond transaction patterns of life insurers in response to movements in long-term interest rates. Specifically, I expect life insurers to rebalance their bond portfolios towards riskier bonds following long-term interest rate increases.

To study changes in insurers' portfolio demand, I first construct the share of their bond purchases in the high-yield market,

$$(\text{Purchase Share})_t^{\text{HY}} = \frac{\text{Purchase}_t^{\text{HY}}}{\text{Purchase}_t^{\text{Total}}}.$$

Here $\text{Purchase}_t^{\text{HY}}$ is life insurers' purchases of high-yield (NAIC 3-6) bonds in the 3-month period after month t ,³⁰ $\text{Purchase}_t^{\text{Total}}$ is the life insurers' purchases of all bonds during the same period, and $(\text{Purchase Share})_t^{\text{HY}}$ measures the fraction of insurers' bond purchases in HY segments.

²⁸The estimates for β can be found in Internet Appendix Table F.4. Post-GFC, the β estimate is positive, suggesting that credit spreads move in the same direction as long rates for bonds not held by insurers.

²⁹Column 3 in [Table A.5](#) further shows that the results remain the same after controlling for bonds' callability and moneyness, as in [Section 3.2](#). The last column in [Table A.5](#) shows that the results are robust to controlling for duration-time fixed effects, as in regression (11).

³⁰After constructing the bond transaction records, I connect them to the WRDS Bond Returns data set for credit rating information.

I then use the following empirical specification to test the hypothesis that life insurers rebalance towards riskier bonds after increases in the long-term interest rate,

$$\Delta(\text{Purchase Share})_{t,h}^{\text{HY}} = \alpha + \beta_h \cdot \Delta y_t^{(10)} + \text{Year FE} + \varepsilon_t. \quad (17)$$

In this specification, $\Delta y_t^{(10)}$ is the change in the 10-year Treasury yield from month t to month $t + 3$. The outcome variable is the change in the HY purchase share from month t to $t + h$. The coefficient β_h measures how much life insurers shift their bond purchases towards HY bonds over the next h months following a 1% rise in the 10-year Treasury yield. As in [Section 3.3](#), I replace $\Delta y_t^{(10)}$, the total change in the long rate, with the changes in the long rate around FOMC meetings (cumulated to the same period), in the specification above,

$$\Delta(\text{Purchase Share})_{t,h}^{\text{HY}} = \alpha + \beta_h \cdot \left(\Delta y_t^{(10)} \Big|_{\text{FOMC}} \right) + \text{Year FE} + \varepsilon_t. \quad (18)$$

[Figure 9](#) shows the estimated coefficients from regressions (17) and (18). After an increase in the 10-year yield, life insurers tilt their bond portfolio towards riskier bonds, again consistent with the duration mismatch channel.³¹ The effects are large and significant, while the regression using FOMC shocks shows noisier but larger responses. After a 1% increase in the 10-year Treasury yield, the composition of life insurers' bond purchases shifts by 5% towards the HY segment over the next 6 months, which amounts to 71% of the average HY purchase share over this period (7.06%, see [Table A.3](#), panel A). Meanwhile, it is worth noting that the impact of the long rate on insurers' bond purchases is gradual and can take up to 6 months to fully materialize. This suggests that the bond market is forward-looking, and price impacts can arise more quickly due to anticipatory behavior by market participants, as shown in [Figure 4](#).

Next, I investigate whether the portfolio rebalancing pattern also holds in the cross section of life insurers, that is, whether insurers with greater exposure to interest rates adjust their bond purchases more strongly. While identifying exogenous variations in insurers' risk exposures has proven challenging, the existing literature (e.g., [Koijen and Yogo 2022](#); [Ellul et al. 2022](#); [Sen 2023](#)) has revealed that a particular type of product sold by life insurers—variable annuities (VAs)—has extremely high duration and convexity due to its minimum return guarantees and embedded options. Consequently, insurers

³¹Similar to findings in this paper, [Fang and Xiao \(2025\)](#) also show that insurers tilt their portfolio towards lower-quality bonds when interest rates fall.

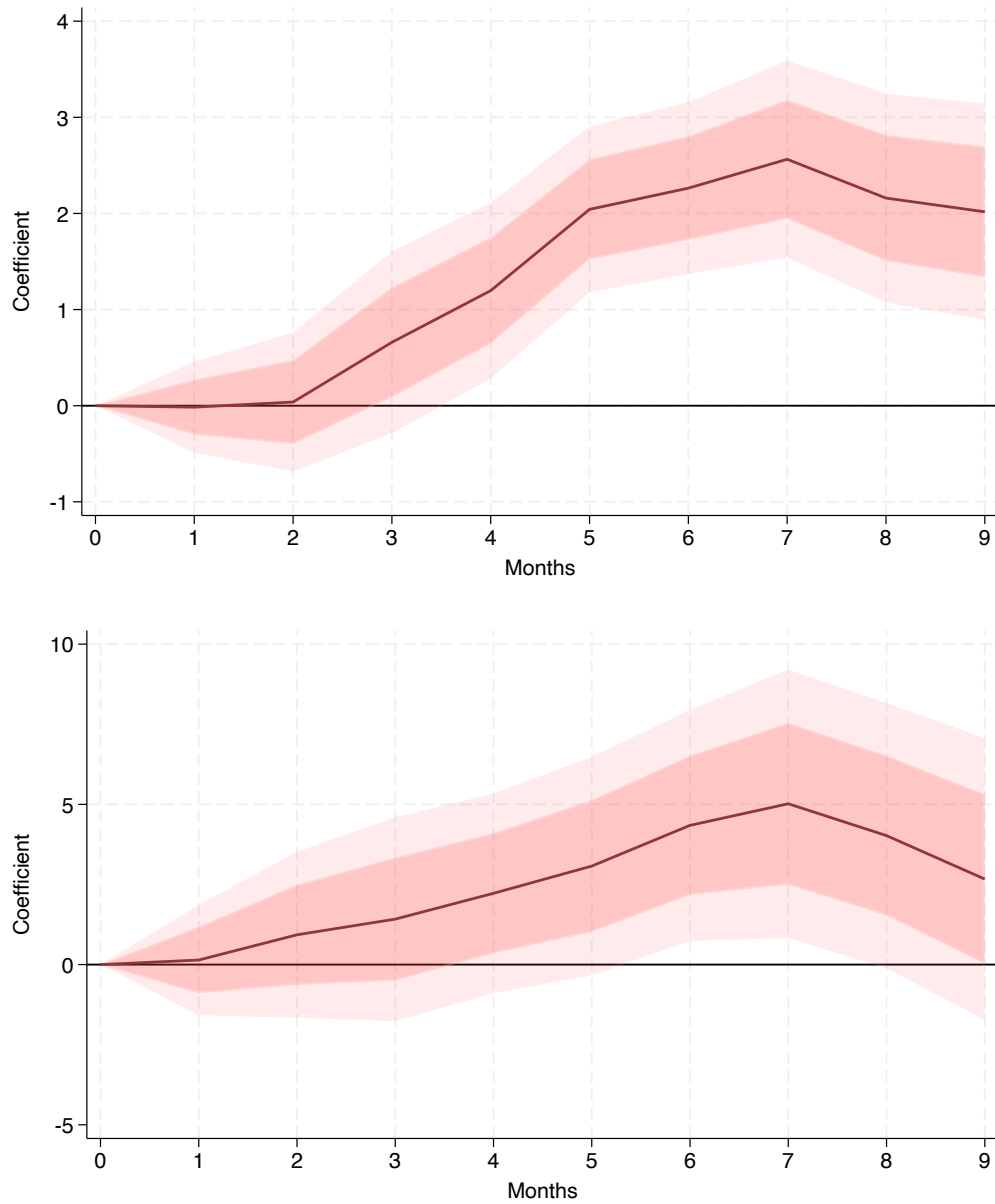


Figure 9. Long rates and the composition of insurers' bond purchases

The top panel shows the coefficients estimated from regression (17). The bottom panel shows the coefficients estimated from regression (18). Year fixed effects are controlled. The shaded areas indicate 68% and 90% confidence intervals based on robust standard errors.

that issued large quantities of VAs pre-GFC ended up facing more severe duration mismatch post-GFC. In light of the link between VA liabilities and insurers' risk exposure, I rerun the previous regression at the individual insurer level, further adding an interaction term between $\Delta y_t^{(10)}$ and insurers' VA liability shares in 2009, the beginning of the post-GFC sample,³²

$$\Delta(\text{Purchase Share})_t^{\text{HY}} = \alpha_j + \alpha_t + \beta \cdot (\text{VA Share})_{j,2009} \cdot \Delta y_t^{(10)} + \Gamma \mathbf{X}_{jt} + \varepsilon_{jt}. \quad (19)$$

Table 7 shows the results from regression (19). I have added insurer and time fixed effects to further control for any factors that are insurer- or time-specific. To alleviate concerns that VA insurers might be larger and more constrained, I further control for the interactions between insurer size and the RBC ratio and $\Delta y_t^{(10)}$. The results show that insurers with stronger interest rate exposures (i.e., those that issued more VAs) exhibited larger portfolio rebalancing towards HY bonds following long rate increases. A one-standard-deviation increase in the insurer's VA share is associated with a 0.94% larger response of the HY purchase share (over 14% of the 6.51% average HY purchase share across all insurers, as shown in Table A.3, panel B). The results support the hypothesis that the duration mismatch of life insurers can drive credit spread dynamics by affecting insurers' bond demand.

Table 7. Variable annuity exposure and insurer bond purchases

	$h = 3m$	$h = 6m$
$(\text{VA Share})_{j,2009} \times \Delta y_t^{(10)}$	0.225*** [4.140]	0.943*** [9.182]
Insurer FE	✓	✓
Time FE	✓	✓
R^2	.022	.036
Observations	27,305	23,582

This table shows the coefficients estimated from regression (19) with insurer fixed effects and time fixed effects, controlling for insurer log net assets and Authorized Control Level RBC ratio, both interacted with $\Delta y_t^{(10)}$. The variable $(\text{VA Share})_{j,2009}$ is standardized. Net assets and the RBC ratio are reported annually, and I use their values as of the end of the previous year. The variable h denotes the horizon of change for $\Delta(\text{Purchase Share})_t^{\text{HY}}$. The t -statistics shown in brackets are based on standard errors clustered at the insurer level. * $p < .1$; ** $p < .05$; *** $p < .01$.

In order to assess the magnitude and quantitative significance of insurers' portfolio

³²I adopt the same definition of the VA liability share as in Kojien and Yogo (2022).

rebalancing, I adopt the following specification to estimate the difference in the quantities of bonds purchased across credit ratings

$$Purchase_t^{NAIC\ k} = \alpha_t + \sum_k \beta_k \cdot \mathbf{1}_{\{NAIC\ k\}} \cdot \Delta y_t^{(10)} + \varepsilon_t. \quad (20)$$

The dependent variable $Purchase_t^{NAIC\ k}$ is life insurers' purchases in NAIC k bonds. The time fixed effect α_t absorbs the average purchases in IG bonds, and the coefficient β_k measures the amount of insurers' purchases in NAIC k that is in excess of their purchases of IG bonds, following a 1% positive increase in the 10-year rate. The results, presented in Table 8, again verify that insurers increase their purchases of HY bonds relative to IG bonds following increases in the long rate.

Table 8. Long rates and insurers' bond purchases

	Pre-2007	Post-2009
$(NAIC\ 3) \times \Delta y_t^{(10)}$	1.236 [0.279]	5.803** [2.018]
$(NAIC\ 4) \times \Delta y_t^{(10)}$	1.114 [0.243]	6.565** [2.089]
$(NAIC\ 5-6) \times \Delta y_t^{(10)}$	1.247 [0.254]	6.955** [2.117]
Time FE	✓	✓
R^2	.04	.038
Observations	270	585

This table shows the coefficients estimated from regression (20). NAIC groups 5 and 6 are combined, as the market size for NAIC 6 is too small to estimate separately. The t -statistics shown in brackets are based on robust standard errors. * $p < .1$; ** $p < .05$; *** $p < .01$.

In the following, I perform simple back-of-the-envelope computations based on Table 8 to gauge the effect of insurers' bond demand on credit spreads. As implied by Table 8, after a 1% rise in the long rate, insurers purchase $5.803 + 6.565 + 6.955 = \19.323 bn of high-yield (NAIC 3-6) bonds in excess of their purchases of investment-grade bonds over the next 3 months. The average total outstanding amount of high-yield bonds in WRDS Bond Returns is \$727.6 bn in my sample (Table A.2), 92.48% of which (\$672.85 bn) is held by investors other than life insurers.³³ Therefore, the excess purchases of HY

³³As Table A.2 shows, 80.9% of HY bonds have positive insurer holdings, among which insurers hold

bonds amount to 2.87% of the HY holdings of all other investors. [Darmouni, Siani, and Xiao \(Forthcoming\)](#) estimate the bond demand elasticity of active mutual funds to be 0.75. Suppose that life insurers mainly trade against mutual funds in the HY market. Then, the price impact of the excess purchases would be about $2.87\%/0.75 = 3.83\%$. Because the average duration of HY corporate bonds in WRDS Bond Returns is 4.45 years, a 3.83% price impact could translate to a $3.83\%/4.45 = 0.86\%$ change in the credit spread between IG and HY bonds.

Next, I compare the 0.86% change in the HY-IG spread to its empirical counterpart. In my sample, the share of NAIC 3, 4, 5, and 6 bonds in the HY market is 54.5%, 35.2%, 9.7%, and 0.95%, respectively, and the share of NAIC 2 among IG bonds is 44.3%. The estimates shown in [Figure 3](#) then imply that the response of the average HY-IG spread to a 1% increase in the 10-year rate is

$$\begin{aligned}
& \Delta(\text{HY-IG Spread}) \\
&= \Delta(\text{Average HY Yield}) - \Delta(\text{Average IG Yield}) \\
&= \sum_{k=3}^6 (\text{NAIC } k \text{ Share in HY}) \Delta y^{\text{NAIC } k} - \sum_{k=1}^2 (\text{NAIC } k \text{ Share in IG}) \Delta y^{\text{NAIC } k} \\
&= \sum_{k=3}^6 (\text{NAIC } k \text{ Share in HY}) \Delta (y^{\text{NAIC } k} - y^{\text{NAIC } 1}) - (\text{NAIC } 2 \text{ Share in IG}) \Delta (y^{\text{NAIC } 2} - y^{\text{NAIC } 1}) \\
&= 54.5\%(-0.75\%) + 35.2\%(-1.38\%) + 9.7\%(-3.69\%) + 0.95\%(-10.72\%) - 44.3\%(-0.13\%) \\
&= -1.3\%.
\end{aligned}$$

Therefore, the 0.86% decline in the HY-IG spread derived from the back-of-the-envelope calculation is able to account for the majority (i.e., around 2/3) of the 1.3% empirical comovement shown in [Figure 3](#).³⁴

The back-of-the-envelope calculations rely on two simplifying assumptions: (1) the purchases of investment-grade bonds are zero,³⁵ and (2) the average demand elasticity

9.3% on average. Thus, noninsurers hold approximately $(1 - 80.9\%) + 80.9\%(1 - 9.3\%) = 92.48\%$ of HY bonds in total.

³⁴It is likely that the duration mismatch channel of life insurers does not explain the entirety of the comovement between long rates and credit spreads. The unexplained variations in credit spread dynamics might be driven by other factors, such as the duration mismatch of other investors (e.g., pension funds).

³⁵This assumption likely biases the inferred yield impacts downwards. Since investment-grade bonds have larger durations and amounts outstanding than high-yield bonds, the same purchases will have a smaller yield impact on investment-grade bonds. Hence, considering purchases in NAIC 1 and 2 would likely increase the magnitude of the response in the HY-IG spread.

of the corporate bond market is the same as the elasticity of active mutual funds. While precision can potentially be improved, these results provide approximations of the plausible magnitudes by which life insurers' bond purchases can move bond yields, and they are very much in line with the main empirical estimates.

5 Conclusion

Following the 2007–2008 Financial Crisis, U.S. life insurers faced a severe duration mismatch, and their net worth declined sharply as long-term interest rates fell. In this paper, I investigate the duration mismatch channel, in which declines in long rates squeeze life insurers' balance sheet equity and thereby increase their effective risk aversion, leading to higher equilibrium credit spreads. To provide empirical support for the duration mismatch channel, I document a shift in the comovement between the long-term interest rate and corporate bond credit spreads. In particular, declines in long-term interest rates led to large increases in credit spreads only after the Financial Crisis, when life insurers began to face negative duration gaps. I further establish that this comovement is more pronounced in bonds held by life insurers, using an exogenous discontinuity in the ownership structure of bonds. The duration mismatch channel can have significant implications for the bond market and monetary policy transmission, given the long-run trend of falling interest rates and the increasing prevalence of unconventional monetary policies (e.g., central bank asset purchases and forward guidance) that specifically target long-term interest rates.

Appendices

Appendix A Supplementary Figures and Tables

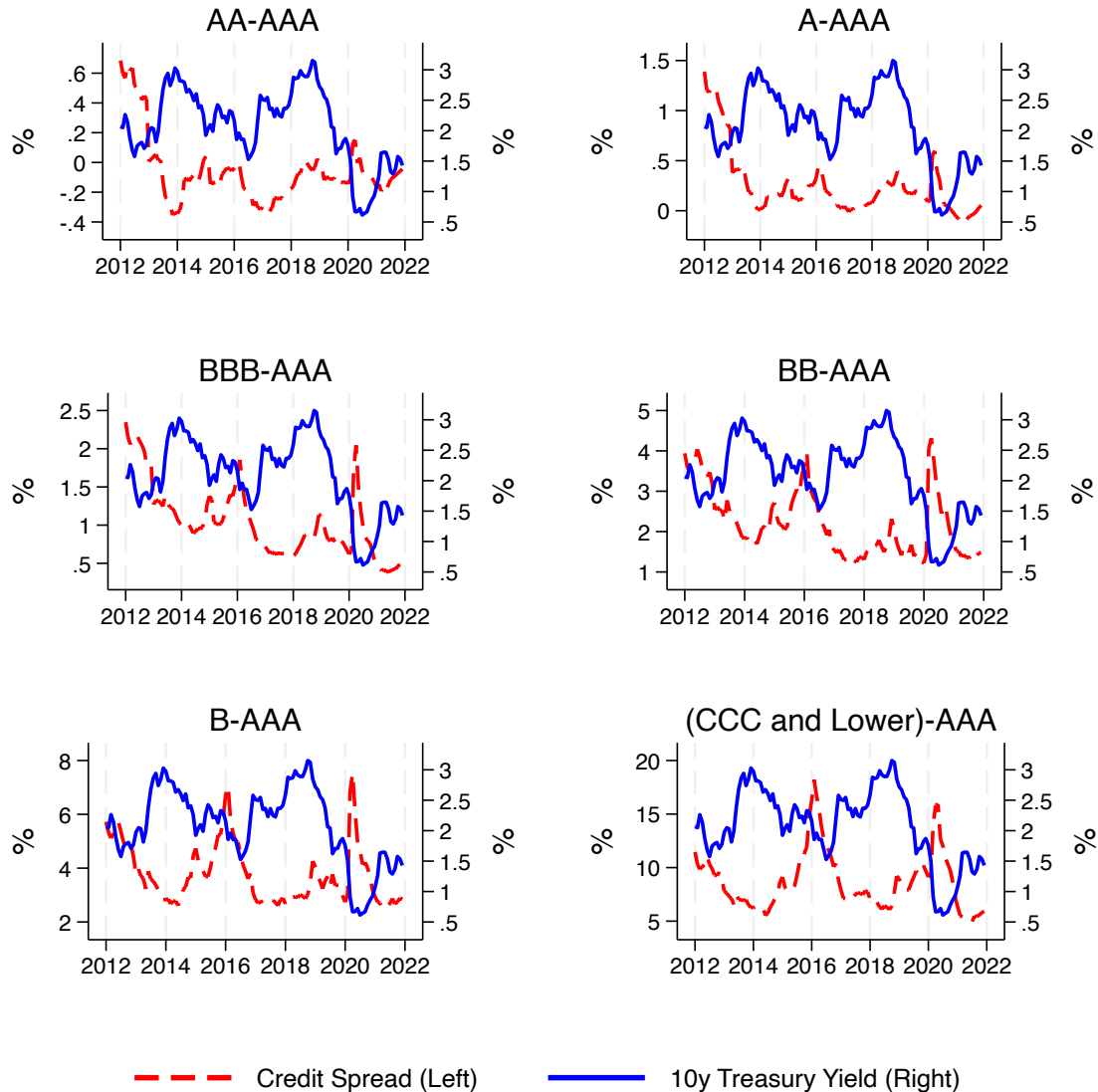


Figure A.1. The 10-year Treasury yield and bond credit spreads (post-GFC)

This figure plots the average monthly 10-year Treasury yield and the spreads of corporate bond indices relative to the index of AAA bonds. The 10-year Treasury yield is shown as the solid blue line and corresponds to the y-axis on the right-hand side. The credit spreads are shown as dashed red lines and correspond to the y-axis on the left-hand side.

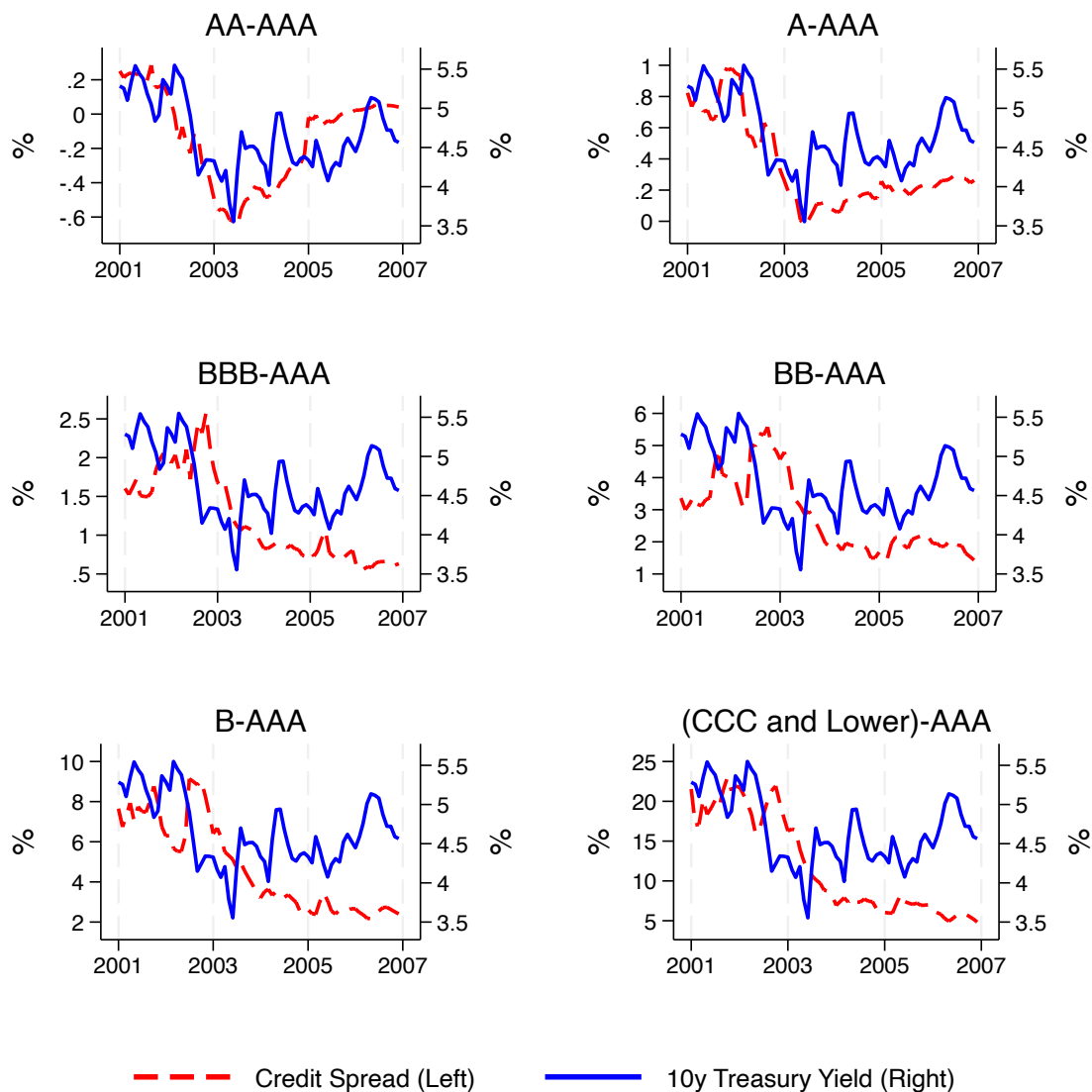


Figure A.2. The 10-year Treasury yield and bond credit spreads (pre-GFC)

This figure plots the average monthly 10-year Treasury yield and the spreads of corporate bond indices relative to the index of AAA bonds. The 10-year Treasury yield is shown as the solid blue line and corresponds to the y-axis on the right-hand side. The credit spreads are shown as dashed red lines and correspond to the y-axis on the left-hand side.

Time Period: 2010-2022 (excl. Mar 2020)

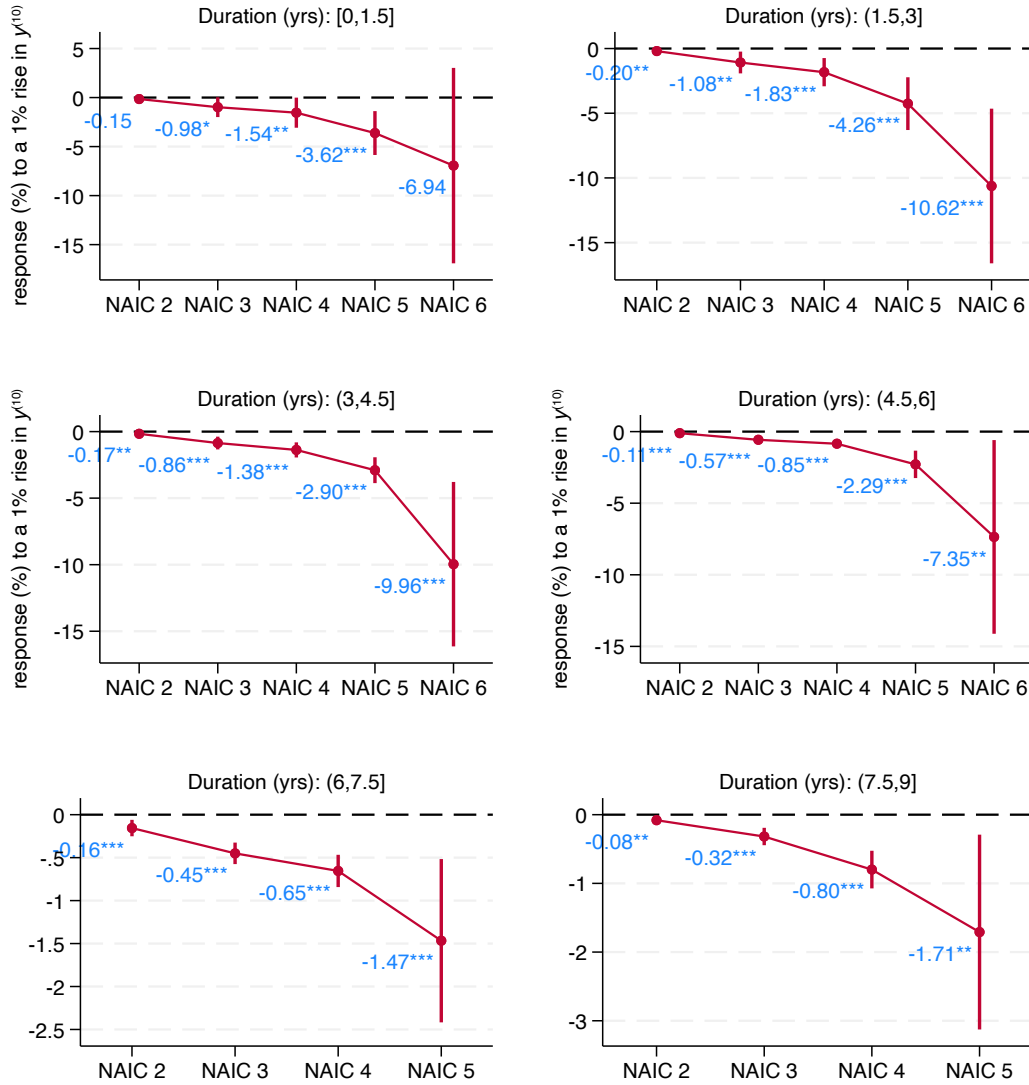


Figure A.3. Credit spread responses in different duration groups (post-2009)

This figure plots the coefficients β_k estimated from regression (11), controlling for the trading volume, outstanding amount, time to maturity, credit ratings, coupon rate, and coupon frequency of each bond, and the recent default rate of each NAIC category, all interacted with $\Delta y_t^{(10)}$. Both corporate bond yields and the Treasury yield are in percentage points. The different panels estimate coefficients for bonds in different duration segments. The NAIC 6 group is omitted in the last two panels because there are too few NAIC 6 bonds with a sufficiently long duration. The statistical significance is based on standard errors clustered at the issuer and year-month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

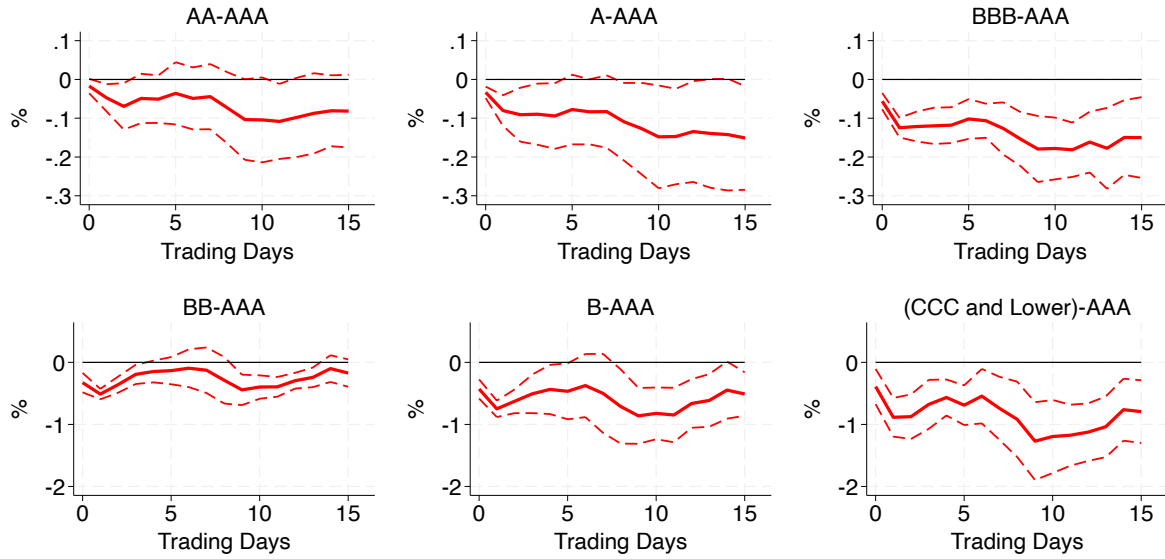


Figure A.4. Impulse responses of credit spreads (2010–2022)

This figure plots the cumulative responses of credit spreads to a 1% increase in the 10-year Treasury yield and 90% confidence intervals based on Newey-West standard errors with a maximum lag of 30 trading days. I now include changes in the 1-month Treasury yield and average CDS spreads around 2-day FOMC windows as controls. In particular, the regression specification is

$$\begin{aligned}
 Spread_{t+h}^k - Spread_{t-1}^k = & \alpha_h + \beta_h \left(\Delta y_t^{(10)} \Big|_{\text{FOMC}} \right) \\
 & + \gamma_h \left(\Delta y_t^{(1m)} \Big|_{\text{FOMC}} \right) \\
 & + \delta_h \left(\Delta(\text{CDS Spread})_t^{\text{Rating } k-AAA} \Big|_{\text{FOMC}} \right) + \varepsilon_{t,h}.
 \end{aligned}$$

Here, $(\text{CDS Spread})_t^{\text{Rating } k-AAA}$ is the difference between the average CDS spread of rating- k bonds and AAA bonds. $\left(\Delta y_t^{(1m)} \Big|_{\text{FOMC}} \right)$ denotes the changes in the 1-month Treasury yield around 2-day FOMC windows. $\left(\Delta(\text{CDS Spread})_t^{\text{Rating } k-AAA} \Big|_{\text{FOMC}} \right)$ denotes the changes in the CDS spread around 2-day FOMC windows.

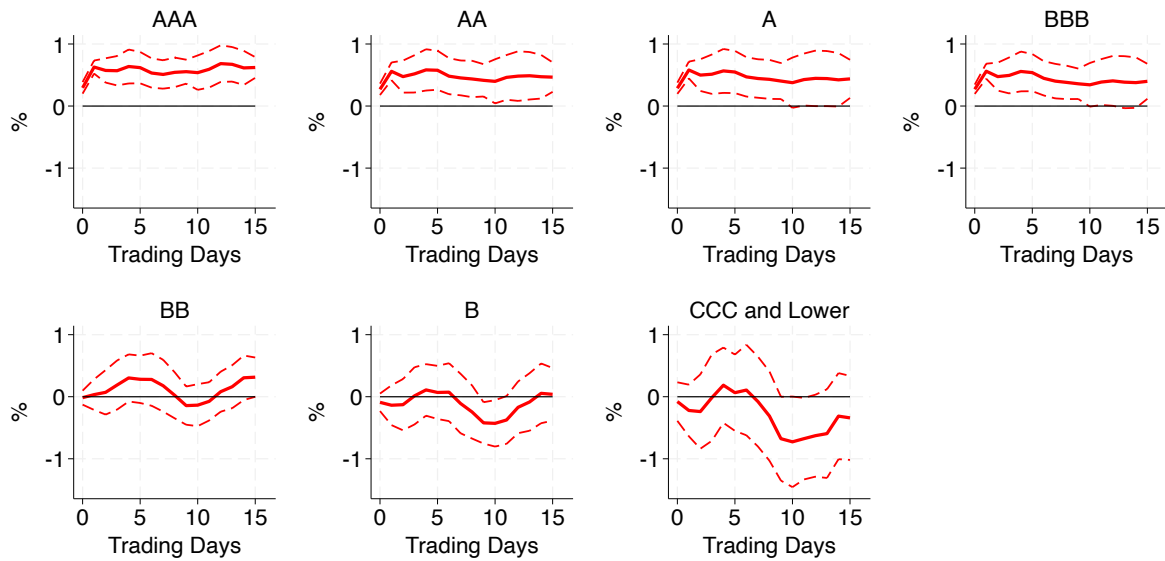


Figure A.5. Impulse responses of yield indices (2010–2022)

This figure plots the cumulative responses of yield indices to a 1% increase in the 10-year Treasury yield and 90% confidence intervals based on Newey-West standard errors with a maximum lag of 30 trading days.

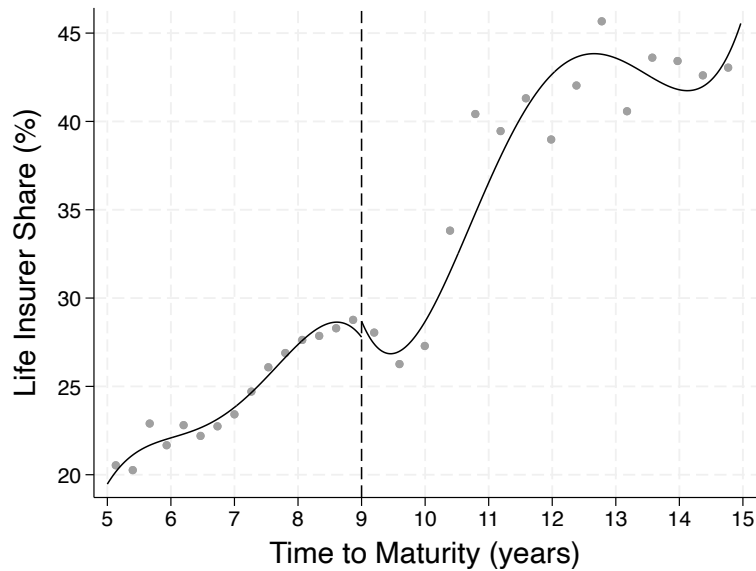


Figure A.6. Discontinuity in investor composition (cutoff = 9 years)

This figure shows a bin scatter plot of corporate bonds' life insurance ownership share (%) and their time to maturity. The vertical line indicates the maturity threshold of 9 years. The figure also shows the best-fit fourth-order polynomials for observations on each side of the threshold.

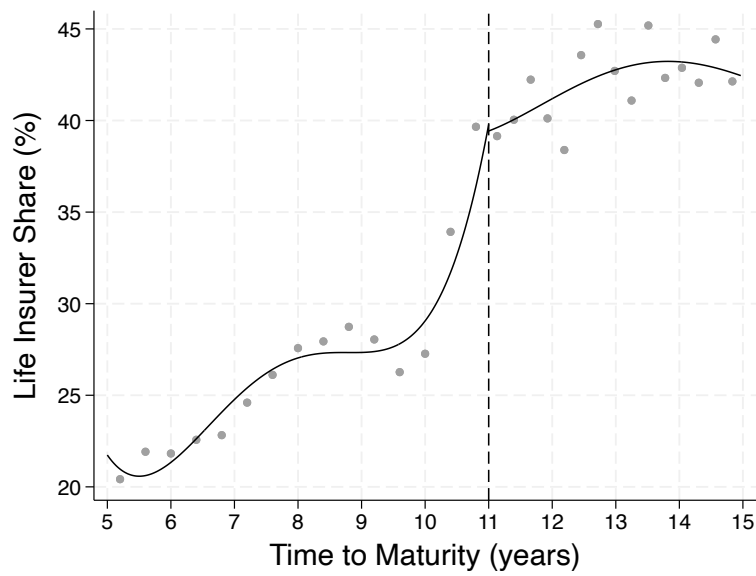


Figure A.7. Discontinuity in investor composition (cutoff = 11 years)

This figure shows a bin scatter plot of corporate bonds' life insurance ownership share (%) and their time to maturity. The vertical line indicates the maturity threshold of 11 years. The figure also shows the best-fit fourth-order polynomials for observations on each side of the threshold.

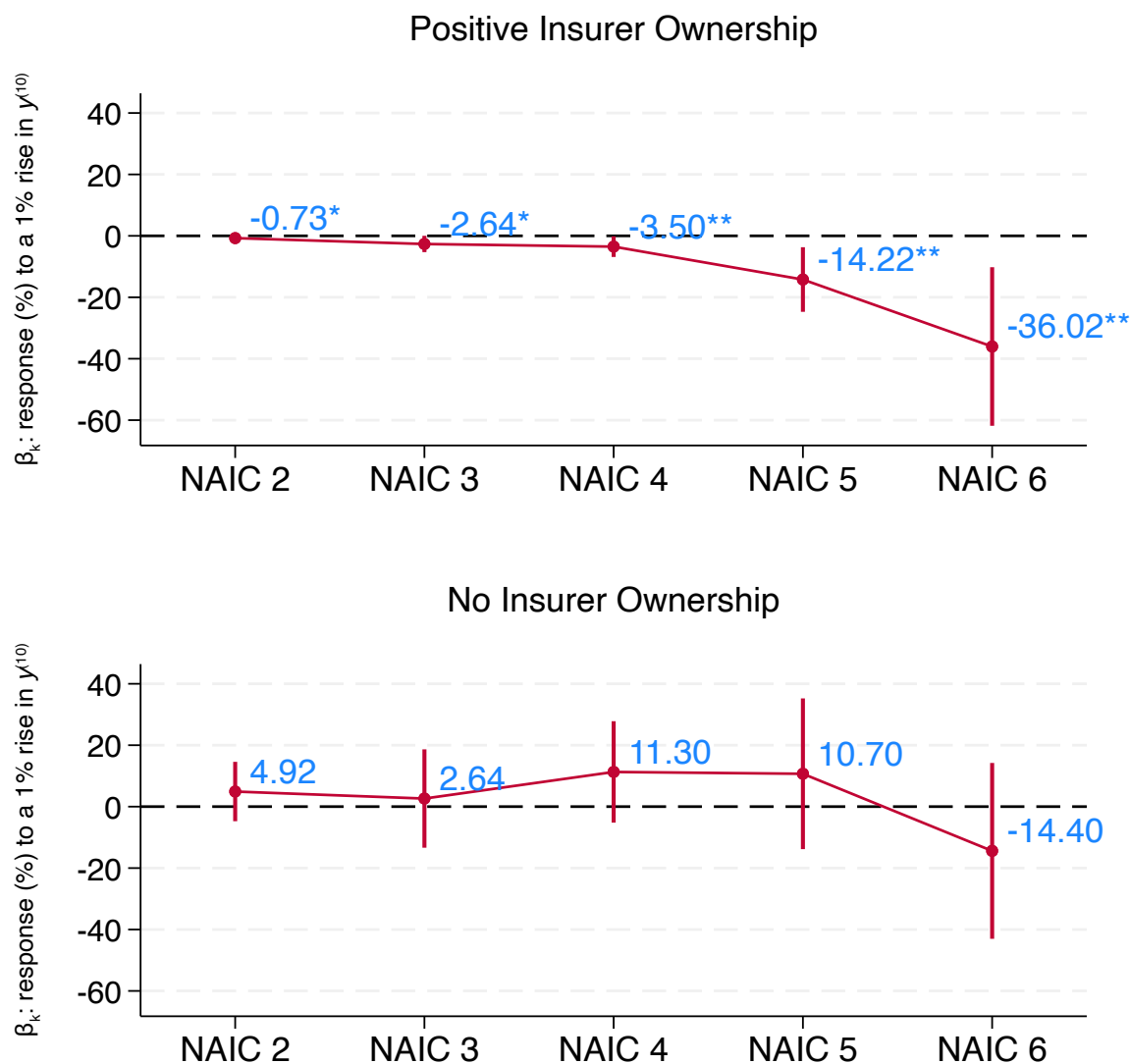


Figure A.8. Credit spread responses by life insurance ownership

This figure plots the coefficients β_k estimated from regression (11) for the period of 2010–2019, controlling for the trading volume, outstanding amount, time to maturity, credit ratings, coupon rate, and coupon frequency of each bond, and the recent default rate of each NAIC category, all interacted with $\Delta y_t^{(10)}$. Both corporate bond yields and the Treasury yield are in percentage points. The top panel shows estimates for the subsample of bonds with life insurance ownership. The bottom panel shows estimates for the subsample of bonds without life insurance ownership. The statistical significance is based on standard errors clustered at the issuer and year-month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

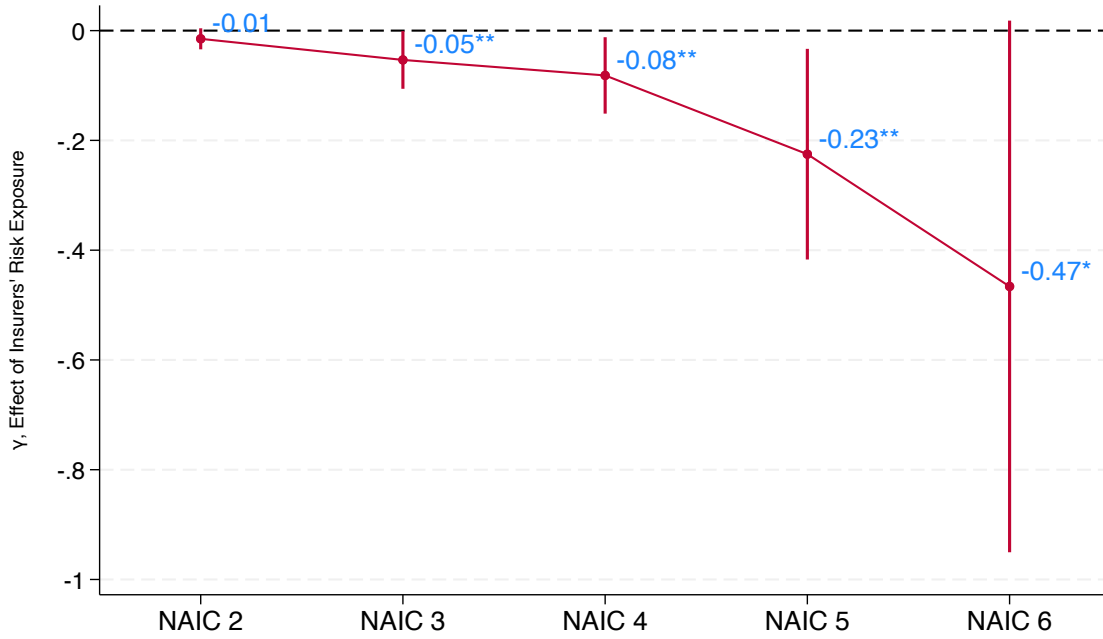


Figure A.9. Insurers' risk exposure and credit spread dynamics

This figure plots the coefficients γ_k estimated from regression (15), controlling for the trading volume, outstanding amount, time to maturity, credit ratings, coupon rate, and coupon frequency of each bond, and the recent default rate of each NAIC category, all interacted with $\Delta y_t^{(10)}$. Both corporate bond yields and the Treasury yield are in percentage points. The regression is estimated for the period between 2002 and 2019, excluding the GFC (2007–2009). The statistical significance is based on standard errors clustered at the issuer and year-month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

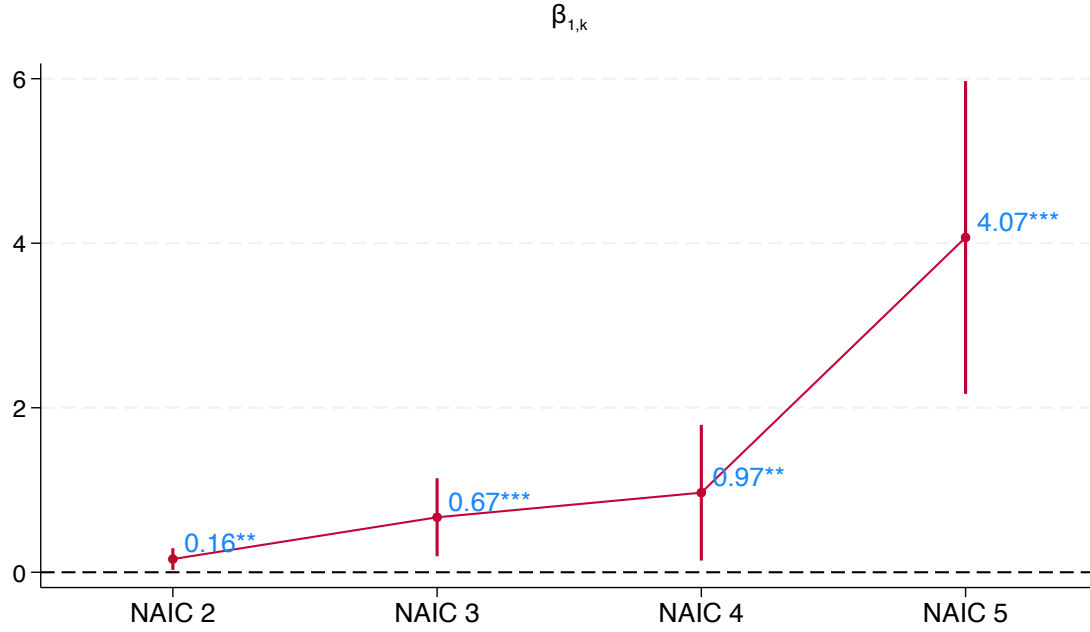


Figure A.10. Changes in credit spread dynamics post-2023

This figure plots the coefficients $\beta_{1,k}$ from the following local projection:

$$\Delta y_{it} = \alpha_i + \alpha_{D(i),t} + \sum_{k=2}^6 \left(\beta_{0,k} + \beta_{1,k} \mathbf{1}_{\{t \geq 2023\}} \right) \cdot \mathbf{1}_{\{\text{NAIC } k\}} \cdot \Delta y_t^{(10)} + \Gamma \mathbf{X}_{it} + \varepsilon_{it}.$$

The regression uses the same controls as in [Figure 3](#). NAIC 6 is omitted due to insufficient observations. The statistical significance is based on standard errors clustered at the issuer and year-month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table A.1. Summary statistics*Summary statistics: Bond yields by NAIC category (unit: %)*

	Mean	SD	P25	P75	N
NAIC 1	2.81	6.50	1.81	4.01	374,340
NAIC 2	3.68	2.60	2.64	4.70	369,611
NAIC 3	5.42	2.51	4.26	6.42	82,052
NAIC 4	7.48	4.11	5.76	8.40	58,839
NAIC 5	13.43	11.97	7.81	14.31	21,877
NAIC 6	36.92	28.59	15.21	51.88	2,891
Total	4.06	5.91	2.43	4.94	909,610

Summary statistics: Credit spreads against NAIC 1 (unit: %)

	Mean	SD	P25	P75	N
NAIC 2 – NAIC 1	0.88	0.27	0.71	0.98	120
NAIC 3 – NAIC 1	2.61	0.61	2.14	3.02	120
NAIC 4 – NAIC 1	4.65	1.23	3.86	5.03	120
NAIC 5 – NAIC 1	10.96	4.11	8.13	12.26	120
NAIC 6 – NAIC 1	33.51	11.67	25.41	40.08	120

Summary statistics: Bond-level variables

	Mean	SD	P25	P75	N
Duration (years)	6.17	4.47	2.76	8.19	907,447
Time to Maturity (years)	9.46	9.83	3.05	11.29	915,733
Outstanding (M)	630.78	634.14	250.00	750.00	916,439
Total Par Value Volume (M)	51.94	184.00	2.42	46.22	916,788
Coupon Rate (%)	5.19	2.14	3.70	6.62	916,785
Coupon Frequency	2.01	0.73	2.00	2.00	916,788

The tables show summary statistics of bond yields, credit spreads, and bond characteristics from WRDS Bond Returns for the period of 2010–2019.

Table A.2. Summary statistics: Insurance ownership

	$\mathbf{1}\{\varphi_{it}^{\text{Ins}} > 0\}$	$\mathbb{E}_t[\varphi_{it}^{\text{Ins}} \varphi_{it}^{\text{Ins}} > 0]$	$\max \varphi_{it}^{\text{Ins}}$	Amount Outstanding (\$ bn)
NAIC 1	88.7%	29.4%	100%	2,268.3
NAIC 2	93.6%	31.8%	100%	1,821.2
NAIC 3	89.3%	13.0%	98.0%	381.3
NAIC 4	79.5%	5.6%	90.9%	254.0
NAIC 5	58.8%	3.4%	75.7%	80.2
NAIC 6	31.5%	2.4%	64.9%	12.1
NAIC 1-2	91.0%	30.5%	100%	4,089.5
NAIC 3-6	80.9%	9.3%	98.0%	727.6

This table summarizes life insurers' bond ownership in WRDS Bond Returns between 2010 and 2019. The first column shows the average fraction of bonds with positive insurer holdings. The second column shows the average life insurance ownership share in bonds with life insurance ownership. The third column shows the maximum life insurer share observed in each category. The last column shows the average total outstanding amount of each category.

Table A.3. Summary statistics: HY purchase shares*A: Aggregate life insurance sector*

	Mean	SD	P25	P75	N
HY Purchase Share (%)	7.06	2.40	4.94	8.59	117

B: Individual insurers

	Mean	SD	P25	P75	N
HY Purchase Share (%)	6.51	14.99	0.00	6.29	34,318

The tables show summary statistics of $(\text{Purchase Share})_t^{\text{HY}}$, the 3-month HY share of net purchases, for the period of 2010–2019.

Table A.4. RDD regressions: Robustness checks

	Bandwidth (years)					
	1.5	1.6	1.7	1.8	1.9	2.0
γ	−9.701* [−2.037]	−11.806*** [−3.688]	−10.466*** [−4.011]	−9.404** [−2.675]	−8.774** [−2.912]	−6.694* [−2.130]
β	4.498 [1.668]	5.565** [3.139]	4.941*** [3.368]	4.282* [2.076]	3.940* [2.237]	2.787 [1.483]
Time FE	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓
First-stage F	84.266	85.269	105.168	103.768	125.748	126.722
Observations	6,273	6,721	7,078	7,684	8,096	8,636

The table shows the β and γ estimated from regression (16) under different choices of bandwidth. The associated t -statistics are reported in brackets.

Table A.5. RDD regressions: Robustness checks

	Subsamples		Callability	Duration
	(1)	(2)	(3)	(4)
γ	-11.366*** [-3.416]	-10.729** [-2.266]	-11.197** [-2.410]	-6.491** [-2.266]
<i>Fixed-Price Call Option</i> $\times \Delta y_t^{(10)}$			-0.786* [-1.920]	
<i>Make-Whole Call Option</i> $\times \Delta y_t^{(10)}$			-7.086** [-2.728]	
<i>Hybrid Call Option</i> $\times \Delta y_t^{(10)}$			-6.845** [-2.786]	
<i>Moneyness</i> $\times \Delta y_t^{(10)}$			0.079** [3.230]	
Time FE	✓	✓	✓	
Duration-Time FE				✓
Controls	✓	✓	✓	✓
First-stage F	91.060	18.887	41.649	100.878
Observations	7,790	2,751	5,624	7,729

This table shows the coefficients estimated from regression (16), with the instrument, controlling for the trading volume, time to maturity, size, coupon rate and frequency, credit rating for each bond, and the recent default rate for each NAIC category, all interacted with $\Delta y_t^{(10)}$, and the bonds' insurer ownership share. The insurance ownership is normalized by its standard deviation. The first column shows the result excluding bonds issued less than a month ago. The second column shows the result for bonds whose maturity is above 10.25 years at issuance. The third column shows the estimates when additional callability controls are included. The fourth column shows the estimates when we control for duration-time fixed effects instead of time fixed effects. The t -statistics shown in brackets are based on standard errors clustered at the issuer and year-month levels. * $p < .1$; ** $p < .05$; *** $p < .01$.

Appendix B Constructing Bond Call Prices

In this section, I describe the method I use to construct the callability and strike prices of a large fraction of corporate bonds in my sample. As in [Ma, Streitz, and Tourre \(2023\)](#), we can categorize bonds into four types based on the type of their call options:

1. Noncallable bonds: bonds without any call option.
2. Fixed-price callable bonds: bonds with fixed call prices (which can potentially vary over time).
3. Make-whole callable bonds: bonds for which the call price is equal to the present value of remaining cash flows, discounted at the U.S. Treasury yield curve plus a make-whole spread.
4. Hybrid bonds: bonds that are initially make-whole but become fixed-price after a certain amount of time.

Due to two data limitations, I can only obtain call prices for a subsample of my data. First, the main source of data on bond callability used in the literature is the “redemption” data set from Mergent FISD, which does not cover the entire sample of WRDS Bond Returns or TRACE. Second, some fixed-price callable bonds have complicated call price schedules. Without the sub-data set “call schedule” of Mergent FISD, I do not observe how the call prices of fixed-price bonds vary over time.

Instead, I obtain the text descriptions of bonds’ call options from the “redemption” data set of Mergent FISD, which indicates if fixed-price bonds are callable at “par” (call prices are equal to the par value) or “prices” (call prices change over time, but the exact prices are not reported). I then keep the following types of bonds in my new sample: (1) noncallable bonds, (2) fixed-price callable bonds whose call price is equal to par, (3) pure make-whole bonds, (4) hybrid bonds whose fixed-price part has a call price equal to par. In other words, I discard bonds with fixed call prices that vary over time due to the lack of data. After this procedure, the new sample includes 66.44% of the observations of my main sample. To my knowledge, [Ma, Streitz, and Tourre \(2023\)](#) is the only recent study that constructs bond-level call prices for the U.S. corporate bond market. In comparison, the final data set of [Ma, Streitz, and Tourre \(2023\)](#) includes 21,508 bonds out of 602,446 bonds issued between 2000 and 2022 (3.57% of the U.S. corporate bond universe).

For make-whole bonds, I use the following formula to compute call prices,

$$(\text{Call Price})_t = \sum_{n=1}^N \frac{CF_{t+n}}{\left[1 + y_t^{(n)} + (\text{make-whole spread})\right]^n}.$$

Here, CF_{t+n} is the bond's cash flow (e.g., coupons and face value) at time $t + n$, $y_t^{(n)}$ is the Treasury yield of maturity n at time t from [Liu and Wu \(2021\)](#),³⁶ and the make-whole spread is obtained from Mergent FISD. Following [Ma, Streitz, and Tourre \(2023\)](#), if the make-whole spread is missing in Mergent FISD, I assume that the spread is 25 bps for investment-grade bonds and 50 bps for high-yield bonds. As in [Ma, Streitz, and Tourre \(2023\)](#), for hybrid bonds, the cash flows include future coupons between now and the first fixed-price call date plus the first fixed-price call price.

Following the literature on derivative pricing, I measure the bonds' moneyness using the following ratio,

$$(\text{Moneyness Ratio})_t = 100 \times \frac{(\text{Bond Price})_t}{(\text{Call Price})_t},$$

which compares the current bond price to the call strike price. Intuitively, the moneyness of an option increases if the market price increases relative to the call price. I have also included noncallable bonds in this new sample, for which I assume the strike price is infinity and the moneyness is zero.

Appendix C Proof of [Proposition 1](#)

Let s_t^n be the credit spread of rating n . The rating-specific holding cost is

$$\gamma^n := a\delta(v^n)^2 + (\zeta^n)^2 > 0.$$

The net supply is homogeneous across ratings

$$\alpha - \beta s_t^n, \quad \alpha > 0, \beta > 0. \tag{C.1}$$

³⁶I replace $y_t^{(n)}$ with the Treasury yield of the longest available maturity (typically 30 years) if n exceeds the maximum available Treasury yield maturity.

The first-order condition implies a linear relation between spreads and portfolio weights (8):

$$s_t^n = \gamma^n w_t^{I,n} \iff w_t^{I,n} = \frac{s_t^n}{\gamma^n}. \quad (\text{C.2})$$

In dollar terms, insurer demand for rating n equals $A_t^I w_t^{I,n}$. Equating insurer dollar demand with outside net supply (C.1) and using (C.2),

$$A_t^I w_t^{I,n} = \alpha - \beta s_t^n \iff \frac{A_t^I}{\gamma^n} s_t^n = \alpha - \beta s_t^n.$$

Solving for the equilibrium spread yields the closed form

$$s_t^n = \frac{\alpha}{\beta + \frac{A_t^I}{\gamma^n}}. \quad (\text{C.3})$$

Differentiate (C.3) w.r.t. A_t^I :

$$\begin{aligned} \frac{\partial s_t^n}{\partial A_t^I} &= -\alpha \frac{1/\gamma^n}{\left(\beta + A_t^I/\gamma^n\right)^2}, \\ \frac{\partial s_t^n}{\partial y_t^T} &= \frac{\partial s_t^n}{\partial A_t^I} \cdot \frac{\partial A_t^I}{\partial y_t^T} = -\frac{\alpha}{\gamma^n} \frac{\frac{\partial A_t^I}{\partial y_t^T}}{\left(\beta + A_t^I/\gamma^n\right)^2} < 0. \end{aligned}$$

This proves the first prediction. Let $\chi^n \equiv 1/\gamma^n$ and define

$$f(\chi) := \frac{\chi}{\left(\beta + A_t^I \chi\right)^2}.$$

From above,

$$\left| \frac{\partial s_t^n}{\partial y_t^T} \right| = \alpha \left(\frac{\partial A_t^I}{\partial y_t^T} \right) f(\chi^n).$$

A direct derivative gives

$$f'(\chi) = \frac{\beta - A_t^I \chi}{\left(\beta + A_t^I \chi\right)^3}.$$

Thus, whenever $A_t^I > \beta / \min_n \{\chi^n\} = \beta \max_n \{\gamma^n\}$, $f'(\chi) < 0$, so f is strictly decreasing in χ . Lower quality implies a larger γ^n and hence a smaller χ^n and a larger $|\partial s_t^I / \partial y_t^T| > |\partial s_t^H / \partial y_t^T|$, which proves the second prediction. The third prediction follows from

$$B_t^n - B_t^m = -\alpha^B (s_t^n - s_t^m).$$

Equilibrium dollar holdings satisfy

$$A_t^I w_t^{I,n} = \alpha - \beta s_t^n.$$

Differentiate w.r.t. y_t^T :

$$\frac{\partial (A_t^I w_t^{I,n})}{\partial y_t^T} = -\beta \frac{\partial s_t^n}{\partial y_t^T} = \beta \frac{\alpha}{\gamma^n} \frac{\frac{\partial A_t^I}{\partial y_t^T}}{(\beta + A_t^I / \gamma^n)^2} > 0.$$

Hence, holdings in each rating increase with y_t^T . For any two ratings n, m ,

$$\frac{\partial}{\partial y_t^T} \left[(A_t^I w_t^{I,n}) - (A_t^I w_t^{I,m}) \right] = \alpha \beta \frac{\partial A_t^I}{\partial y_t^T} [f(\chi^n) - f(\chi^m)].$$

Since f is decreasing, if $\gamma^n > \gamma^m$, then $\chi^n < \chi^m$ and $f(\chi^n) > f(\chi^m)$, so the insurer rebalances towards lower quality as y_t^T rises. This proves the fourth prediction.

All derivatives above are linear in $\partial A_t^I / \partial y_t^T$. A more negative duration gap increases $\partial A_t^I / \partial y_t^T$, proportionally scaling the magnitudes in predictions 1, 3, and 4. This proves the fifth prediction.

Code and Data Availability The replication code and data are available in the Harvard Dataverse at <https://doi.org/10.7910/DVN/VZ3HSC>.

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